



Contents lists available at ScienceDirect

Journal of Financial Stability

journal homepage: www.elsevier.com/locate/jfstabil



The benefits are at the tail: Uncovering the impact of macroprudential policy on growth-at-risk^{☆,☆☆}

Jorge E. Galán

Financial Stability and Macroprudential Policy Department, Banco de España, Alcalá 48, 28014, Madrid, Spain

ARTICLE INFO

Article history:

Received 26 February 2020

Received in revised form

16 November 2020

Accepted 18 November 2020

Available online xxx

JEL classification:

C32

E32

E58

G01

G28

Keywords:

Financial stability

Growth-at-risk

Systemic risk

Macroprudential policy

Quantile regressions

ABSTRACT

I uncover heterogeneous effects of macroprudential policy on GDP growth distribution by bringing together the literature on the impact of macroprudential policy and recent developments on the use of quantile regressions. I identify important benefits of macroprudential policy on the left-tail of the GDP growth distribution which contrast with the negative effects found in previous studies using conditional mean models. These benefits may offset the deterioration on growth-at-risk produced by the build-up of cyclical vulnerabilities and the materialization of financial crises. I also find that the impact of macroprudential policy is dependent on the position in the financial cycle, the type of instrument, and the time elapsed since its implementation. In particular, tightening capital measures during expansions may take up to two years to show evidence of benefits on growth-at-risk, while the positive impact of borrower-based measures is rapidly observed. Conversely, in downturns the benefits of loosening capital measures are more immediate, while those of borrower-based measures are limited. This suggests the importance of timing in macroprudential policy. Overall, this study provides a useful framework to assess the impact of macroprudential policy in terms of GDP growth and to identify the term-structure of specific instruments.

© 2020 Elsevier B.V. All rights reserved.

1. Introduction

The global financial crisis has evidenced the high costs of the accumulation of financial imbalances for the real economy. Aikman et al. (2019a) estimate that financial vulnerabilities built-up during the years previous to the great recession explain around three-quarters of the subsequent output loss in the US. Moreover, they argue that the magnitude of the negative impact could have been significantly reduced by the active use of macroprudential policies. Certainly, macroprudential policy is aimed at increasing the resilience of the financial sector to negative shocks, limiting the

incentives for risk-taking and taming the financial cycle. This enables the prevention and mitigation of severe negative effects from systemic risk on economic growth, which is considered to be the ultimate objective of macroprudential policy (FSB-IMF-BIS, 2011; European Systemic Risk Board (ESRB), 2015).

Against this background, most previous studies have identified benefits of macroprudential policy in various dimensions such as curbing credit and house price growth (Claessens et al., 2013; Cerutti et al., 2017), reducing the probability of systemic crises (Dell'Ariccia et al., 2016), increasing the probability of survival of firms in a crisis (Jiménez et al., 2017), or decreasing the probability of bank default (Altunbas et al., 2018).¹ However, the few studies measuring the impact of macroprudential policy on GDP growth have identified negative effects. Kim and Mehrotra (2018) identify a negative impact of macroprudential policy on output after analysing an aggregation of many different instruments in Asian economies. Richter et al. (2019) find that borrower-based measures

[☆] I thank Ángel Estrada, Javier Mencía and Javier Suárez for their helpful comments and suggestions. I acknowledge the comments of the participants of the research seminar at Banco de España and the International Finance and Banking Society Conference 2019. I also thank my colleagues of the working groups of the European Systemic Risk Board, where previous versions of this study have been discussed, for their very useful suggestions. I especially acknowledge the improvement ideas of two anonymous referees, from which the paper has benefited in a very significant way.

^{☆☆} This paper is the sole responsibility of its author. The views represented here do not necessarily reflect those of the Bank of Spain or the Eurosystem.

¹ See Galati and Moessner (2018) for a literature review of studies on the effects of macroprudential policy. The authors evidence that this literature is still in its infancy and document the lack of studies assessing the impact on ultimate target variables.

have negative effects on output growth over a four-year horizon. [Noss and Toffano \(2016\)](#) and [Bedayo et al. \(2020\)](#) also identify a negative impact of capital measures on GDP growth in the short-run. These negative effects have been associated with the costs of macroprudential policy. Certainly, by reducing credit supply, macroprudential policy would negatively affect private consumption and investment, thereby dampening economic growth. Thus, macroprudential policy would reduce financial stability risk at the expense of lower output growth.

However, if macroprudential policy effectively reduces systemic risk, one could expect these benefits to be observed in a reduction of the downside risk of GDP growth rather than in its conditional mean. In this regard, [Boar et al. \(2017\)](#) find that the countries which are most active in the use of macroprudential policy achieve a reduction in the variance of output growth. The potential simultaneous reduction in the conditional mean and variance of GDP growth suggests that macroprudential policy would affect the shape of its distribution. In this context, quantile regressions, introduced by [Koenker and Bassett \(1978\)](#), offer a flexible method to disentangle heterogeneous effects on the moments of the GDP growth distribution.

Using this method, [Adrian et al. \(2019\)](#) identify that financial conditions explain the skewness of the GDP growth distribution at short horizons and provide new evidence on the underestimation of the downside of GDP growth when using traditional models focused on the conditional mean. In particular, quantile regressions allow the impact on the left tail of the GDP growth distribution to be assessed, which is associated to the concept of growth-at-risk. This method has also been previously used to study the impact of asset prices ([Cecchetti and Li, 2008](#)), systemic risk measures ([Giglio et al., 2016](#)), and productivity shocks ([Loria et al., 2019](#)) on growth-at-risk. Closer to macroprudential policy, [Aikman et al. \(2019b\)](#) use quantile regressions to forecast the GDP growth distribution conditional on the level of banks' capital ratio, and [Duprey and Ueberfeldt \(2020\)](#) and [Brandao-Marques et al. \(2020\)](#) study the complementarity between macroprudential and monetary policy.

In contrast to the negative effects previously identified using conditional mean models, quantile regressions would allow the potential benefits of macroprudential policy in terms of GDP growth to be identified, which would be presumably observed on the left side of the distribution, given the channels through which macroprudential policy acts. Against this background, the aim of this study is to uncover heterogeneous effects of macroprudential policy on the GDP growth distribution by bringing together two strands of literature, one on the impact assessment of macroprudential policy and the other on the use of quantile regressions to identify effects on growth-at-risk. Therefore, this study focuses on the marginal effects of macroprudential policy and the term structure of its impact over the financial cycle rather than forecasting GDP growth.

For this purpose, I extend the work in [Adrian et al. \(2019\)](#) by estimating quantile regressions of future GDP growth up to four years ahead on a set of macrofinancial variables and macroprudential policy. I use a panel data set including information on a large number of macroprudential measures implemented in advanced and emerging economies starting in 1990, which enables unobserved country-specific heterogeneity to be controlled for. Besides the inclusion of financial conditions, which [Adrian et al. \(2019\)](#) find to be a very relevant determinant of the shape of the GDP growth distribution at short-horizons, I also include macrofinancial variables with good mid-term early warning properties of cyclical systemic risk such as credit growth and house price growth. This allows GDP growth in the mid-term to be characterized, when macroprudential policy might have more relevant effects. Moreover, the analysis of the impact over a mid-term horizon provides interesting insights into the speed and persistence of macropruden-

tial policy on the shape of the GDP growth distribution. Certainly, the time required for macroprudential policy to work has not been fully investigated and there are a wide range of results in the few studies analysing this (see [International Monetary Fund \(IMF, 2014; Noss and Toffano, 2016; Jiménez et al., 2017; Poghosyan, 2019](#), for some attempts to identify the impact lag of capital provisions and borrower-based measures).² This can be related to the fact that most of these studies analyse a few measures taken in a single country, assess the impact on different variables of interest, and do not explicitly account for differential effects throughout the financial cycle. In this regard, I differentiate the effects of tightening and loosening policies and their relationship with the position in the financial cycle. In fact, macroprudential policy has countercyclical objectives, and many instruments facilitate this type of utilization. Finally, given that macroprudential policy encompasses diverse types of measures which may differ in the speed and magnitude of their impact on GDP growth, I provide a more detailed analysis of the effects of the two most-used types of macroprudential instruments (i.e. borrower-based and capital measures).

The results of this study uncover important heterogeneous effects of macroprudential policy on the quantiles of the GDP growth distribution. On one hand, the impact on the median and the right tail of the distribution tends to be negative, validating some previous results using conditional mean models ([Kim and Mehrotra, 2018; Richter et al., 2019](#)). On the other hand, I find important benefits of macroprudential policy on the left tail, which may offset the identified increase in the left-skewness produced by the accumulation of cyclical vulnerabilities and the materialization of systemic risks (see [Adrian et al., 2019](#), for similar results of the impact of tighter financial conditions). Moreover, the position in the financial cycle has a significant role in determining the magnitude and speed of the effects of macroprudential policy on the downside risk of GDP growth. In particular, tightening macroprudential policy during expansionary phases has a positive impact in the mid-term, while implementing loosening measures during crises significantly improves growth-at-risk in the short-term. This asymmetric response of growth-at-risk to the direction of macroprudential policy over the cycle is consistent with previous differential effects identified on the mean of credit and house prices ([Claessens et al., 2013; Cerutti et al., 2017; Poghosyan, 2019](#)). However, previous studies usually do not account simultaneously for the direction of the policy and the position in the financial cycle. A further inspection of the impact of specific types of instruments during expansions suggests rapid and persistent positive effects of borrower-based measures on growth-at-risk, while the benefits of implementing capital measures may take around 8 quarters to be observed. Conversely, releasing capital has rapid positive effects when crises materialize; while loosening borrower-based measures has a limited impact. From a policy perspective, these results indicate the importance of timing in macroprudential policy. While capital measures should be taken early enough in the cycle, policy actions to tighten lending standards could be implemented in more advanced stages.

² [IMF \(2014\)](#) find that tightening macroprudential measures in Israel has had an impact on reducing housing credit over the following six months. However, the impact on longer horizons is not assessed. [Noss and Toffano \(2016\)](#) study the case of the UK and find that tightening capital requirements have a maximum impact on credit and GDP just two quarters after the implementation. More recently, [Poghosyan \(2019\)](#) identify that borrower-based measures may take up to three years to become effective in reducing housing prices and credit and that the response is asymmetric depending on the direction of the measures. However, none of these studies explicitly model the relation with the position in the financial cycle. On the other hand, [Jiménez et al. \(2017\)](#) find that releasing provisions during busts has a progressive impact on credit that reach a maximum after two years. Nonetheless, this study focuses on the Spanish experience with the dynamic provisioning system and does not assess the impact of tightening during expansions.

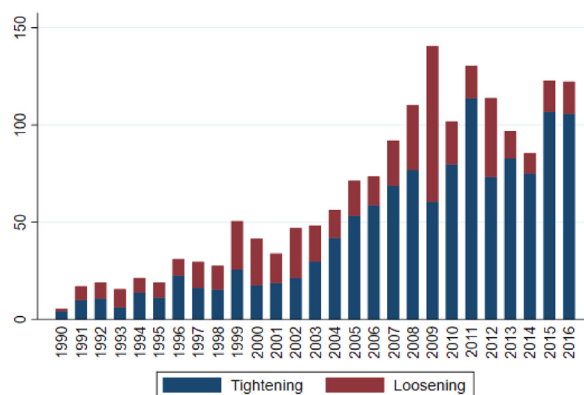


Fig. 1. Use of macroprudential measures by the direction of the policy (1990–2016).
Source: IMF. Own elaboration.

Besides this introduction, the rest of the paper is organized in four additional sections. Section 2 describes the use of macroprudential policy in the last decades and its relationship with the financial cycle. Section 3 presents the relationship between macroprudential policy and GDP growth. Section 4 describes the methodology, the data and the empirical specification. Section 5 analyses the estimation results. Section 6 presents some robustness exercises. Finally, section 7 concludes the paper and discusses some implications for policymakers.

2. The use of macroprudential policy and the financial cycle

The use of macroprudential policy in the last decades has been documented in recent studies using databases that collect information on the macroprudential measures implemented by a large number of national authorities (see Boar et al., 2017; Cerutti et al., 2017; Alam et al., 2019).³ One of the most complete databases is the Integrate Macroprudential Policy database (iMaPP) administered by the International Monetary Fund (IMF) and introduced by Alam et al. (2019), which contains information about macroprudential measures implemented in 134 countries from 1990 to 2016.⁴

Using this database, in Fig. 1 I present the evolution in the use of macroprudential policies over time distinguishing by the direction of the implemented measures. Although during the decade of the 90's and in the run-up to the last financial crisis macroprudential policies began to be actively implemented, mainly in Asian and emerging economies; it was only after the global financial crisis

when the use of macroprudential measures increased significantly. This was mainly due to the active implementation of these policies in advanced economies encouraged by the adoption of new regulations and the consequences of the crisis (see Annex 2 for a description of the use of macroprudential measures distinguishing between advanced and emerging economies). Regarding the direction of the measures, it is observed that most of the implemented policies during the years preceding the global financial crisis were tightening measures, which were taken as a response to the build-up of financial imbalances. Conversely, during the years of the crisis the number of loosening actions increased, as a response on the part of national authorities to the materialization of systemic risk. Then, in recent years tightening measures increased again due to the adoption of new regulatory frameworks and the recovery phase of the financial cycle experienced in many countries.

Thus, a cyclical pattern in the use of macroprudential measures is observed. In Fig. 2, I plot the relationship between some measures of the position in the financial cycle (the financial conditions index (FCI), and the annual growth rates of the credit-to-GDP ratio and house prices) and a measure of the use of macroprudential policy that accounts for its direction. This is the net sum of the implemented macroprudential measures in the last 4 quarters as defined in the categories of the iMaPP (see Section 4.3 for details on the construction of the index). Thus, positive values indicate a net tightening position and negative values indicate a net loosening position. It is observed that national authorities actively respond to credit and house price developments by tightening macroprudential policy, especially during periods where both variables increase simultaneously. In this regard, Jordà et al. (2015) identify that periods of high credit growth precede financial crises, and that their severity is magnified if the previous expansion has been accompanied by large increases in house prices. Therefore, macroprudential authorities monitor developments of the financial cycle in order to take decisions on the implementation of macroprudential policy with the aim of increasing the resilience of the financial sector and mitigating severe negative effects if a financial crisis takes place.

When systemic risk materializes during financial crises, national authorities also act by implementing macroprudential policy but in the opposite direction. That is, by easing conditions that help to mitigate the negative effects of systemic events. Certainly, it is observed that loosening measures are mostly implemented as a response to large increases in the FCI, which encompasses several indicators of market volatility and financial stress.

3. How does macroprudential policy affect output growth?

In the short-run, macroprudential policy acts by curbing excessive credit growth and attenuating financial imbalances. Nonetheless, the ultimate objective of macroprudential policy is to prevent financial crises and to mitigate severe negative effects on the real economy. Thus, in the mid-term the effectiveness of macroprudential policy can be measured in terms of its ability to reduce the occurrence of financial crises and their negative impact on economic growth (see European Central Bank (ECB, 2009; FSB-IMF-BIS, 2011; European Systemic Risk Board (ESRB, 2015; for formal definitions of the ultimate objective of macroprudential policy in terms of economic growth). In this context, macroprudential policy is intended to have a positive impact on economic growth in the mid-term through an increase in the resilience of the financial sector to negative shocks, and a moderation of financial imbalances that could be abruptly corrected during crises with large negative effects on consumption, investment and growth. In this regard, Aikman et al. (2019a) identify that an active use of macroprudential policy would have significantly reduced the large output drop experienced during the global financial crisis.

³ Boar et al. (2017) construct a database of macroprudential measures in 64 countries from 1990 to 2014. That database is based on the information reported in Cerutti et al. (2017); Kuttner and Shim (2016) and Lim et al. (2011). The authors report that emerging economies were especially active in the use of macroprudential policy during the pre-crisis period. Alam et al. (2019) report that around two-thirds of the countries in the sample had implemented at least one macroprudential instrument before the global financial crisis.

⁴ This database integrates information from the Bank of International Settlements (BIS), the European Systemic Risk Board (ESRB), the Financial Stability Board (FSB), IMF's surveys, official communications of national authorities and previous studies. The database classifies macroprudential measures into 17 categories: countercyclical capital buffers (CCyB), capital conservation buffers (CCoB), capital requirements (including risk-weights, minimum requirements and systemic risk buffers), leverage limits, loan loss provisions, limits on credit growth, loan restrictions, limits on foreign currency, limits on Loan-to-Value ratio (LTV), limits on the Debt-to-Income (DTI) and Loan-to-Income (LTI) ratios, tax measures, liquidity requirements, limits on loan-to-deposits ratio (LTD), limits on foreign exchange positions, reserve requirements used for macroprudential purposes (excluding those used for monetary policy), systemically important financial institutions (SIFI) surcharges, and other measures (including stress testing restrictions on profit distributions, and structural measures). See the definition of each category in Annex 1.

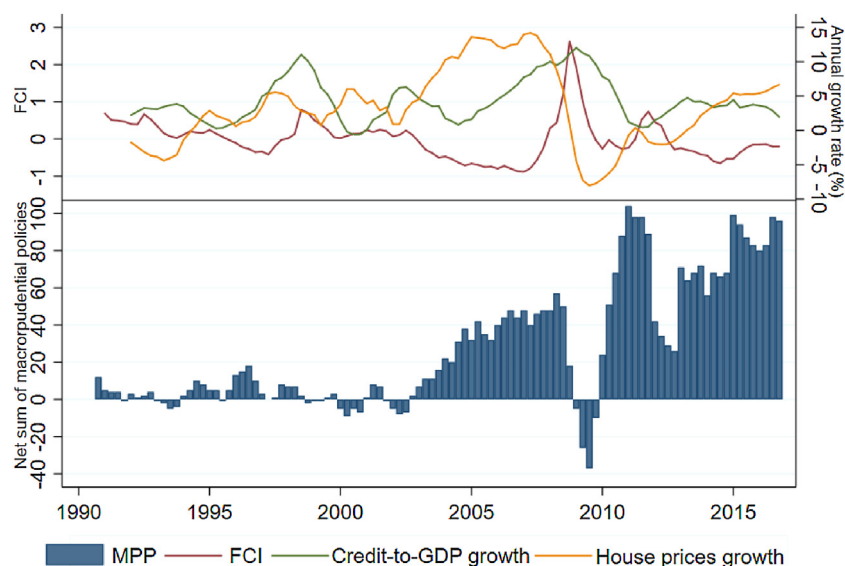


Fig. 2. The financial cycle and the use of macroprudential measures (1990–2016).

Note: The upper part of the left axis represents the value of the FCI. The lower part of the left axis represents the net sum of macroprudential policies. The right axis represents the annual growth rates of the credit-to-GDP ratio and house prices. Data on credit-to-GDP ratio and house prices are obtained from the Bank of International Settlements (BIS). The FCI is obtained from the IMF as well as the information on macroprudential policies. More details on the data are presented in Section 5. MPP stands for macroprudential policy and is represented by an index that is constructed as the net sum of macroprudential policies over the last 4 quarters in the 17 categories defined in the iMaPP (see Section 4.3. for details in the construction of the index). A positive value of this index represents a net tightening position and a negative value represents a net loosening position.

Source: IMF and BIS. Own elaboration.

The countercyclical use of these instruments would also enable positive effects on the economy. Jiménez et al. (2017) identify that loosening countercyclical macroprudential instruments during crises leads to an increase in firms' probabilities of surviving and a lower reduction of employment. These benefits would be transmitted to the real economy through a lower impact and a more rapid recovery of consumption, investment and economic growth.

Nonetheless, macroprudential policy can also have negative effects on economic growth when it is tightened. That is, since macroprudential policy has been found to effectively contain credit growth during expansionary phases of the cycle (Claessens et al., 2013; Cerutti et al., 2017), it would slow investment and private consumption, thereby dampening economic growth. Also, to the extent macroprudential policy curbs asset prices, wealth effects would also reduce consumption and investment. In this regard, Alam et al. (2019) identified that macroprudential policy have negative effects on private consumption in the short-term. Against this background, most of the few recent studies assessing the impact of macroprudential policy on GDP growth have identified negative effects, which are attributable to this channel (Noss and Toffano, 2016; Kim and Mehrotra, 2018; Richter et al., 2019).

In this context, the impact of macroprudential policy on GDP growth depends on the direction of the implemented actions, which are taken as a response to the position in the financial cycle. Although, during financial crises both the business and the financial cycle coincide, they do not always move simultaneously. As has been previously well documented, the financial cycle has a wider amplitude and longer duration than the business cycle (Claessens et al., 2012; Aikman et al., 2015). In particular, the business cycle has been found to have a mean duration of around 5 years, while the length of the financial cycle is at least about 15 years (Drehmann et al., 2010; Galati et al., 2016; Galán and Measuring Credit-to-GDP Gaps, 2019). These differences between the two cycles are evident in the use of macroprudential policy, which primarily responds to the developments of the financial cycle rather than to those of the business cycle. Nonetheless, when both cycles coincide larger vul-

nerabilities are accumulated and the most severe negative effects are observed (Jordà et al., 2015).

The last global financial crisis provides evidence of the impact of macroprudential policy on GDP growth. In Fig. 3, I plot the distribution of the annual GDP growth rate during the global financial crisis and the recovery years by distinguishing between those countries implementing macroprudential policies in the expected direction in the preceding years and those that did not implement any macroprudential measure. It is observed that the shape of the GDP growth distribution differs between the two types of countries. Although there are some differences in the median, the main differences are observed in the left-tail of the distribution. In particular, the distribution of countries that did not implement any policy is highly left-skewed, indicating that the largest contractions of GDP were observed in countries that did not tighten macroprudential policy in the previous four years. This suggests that countries tightening macroprudential policy during the expansionary phase of the financial cycle achieved a financial sector which was more resilient to the negative shock leading, in turn, to a less severe economic contraction when risks materialized. Conversely, those countries easing macroprudential policy during the crisis experienced higher GDP growth during the following four years than those countries that were not actively using macroprudential policy. Again, the difference in the median is less evident than on the left-tail, indicating that countries loosening macroprudential policy during bad times induced a higher and more rapid economic recovery, probably by alleviating the problems of the financial sector, and promoting the flow of credit to the economy. In this regard, IMF (2020) reports that emerging economies entering a global financial shock while being less active in the use of macroprudential policy suffer larger GDP drops than the more active countries.

This suggests that the impact of macroprudential policy on economic growth is heterogeneous across its distribution. Certainly, since the ultimate objective of macroprudential policy is to prevent financial crises and to mitigate their large negative effects on economic growth, it is very likely that macroprudential policy has an important impact on the tail risk of output growth. This would

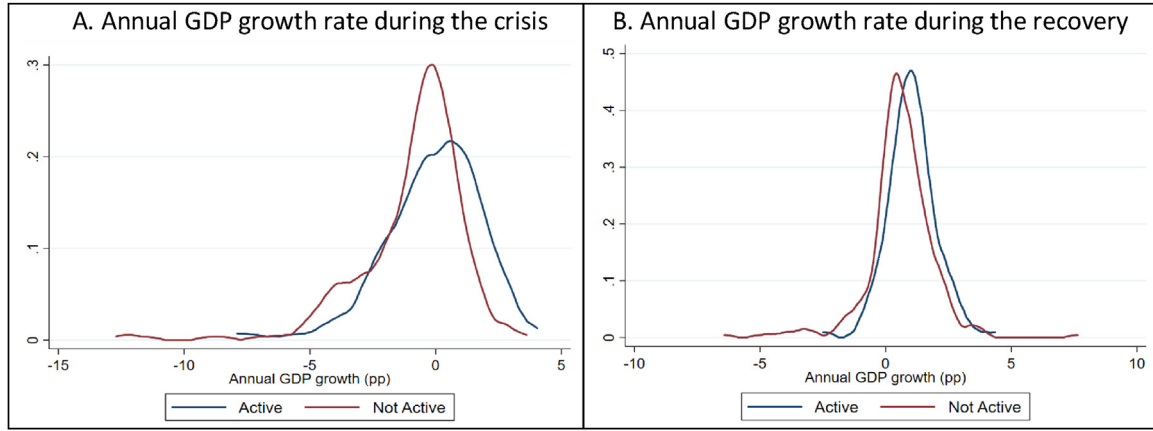


Fig. 3. Annual GDP growth distributions during the global financial crisis and the recovery phase by active and non-active countries in the use of macroprudential policy. Note: The crises years are defined as those two years with the largest average GDP drop in the sample (2008–2009). The recovery period is defined as being from 2010 to 2013. Data on GDP growth is obtained from OECD and the information on macroprudential policies from IMF. More details on the data are presented in Section 5. Source: IMF and OECD. Own elaboration.

be similar to the heterogeneous effect of tighter financial conditions on the GDP growth distribution that was recently identified by Adrian et al. (2019). Thus, in order to disentangle effectively the negative and positive effects of macroprudential policy on economic growth, it would be necessary to account for the direction of the policy, the position in the financial cycle, the time elapsed from the implementation, and the impact on the different moments of its distribution.

4. Methodology

4.1. Panel quantile regressions

Koenker and Bassett (1978) proposed quantile regressions as a useful tool for the identification of differential effects on the distribution of a variable of interest instead of focusing on the conditional mean, which may mask distributional effects. Since then, the methodology has been applied in different fields. In finance the most standard application is the computation of value at risk (Jorion, 2001), which is the computation of the expected loss of a portfolio given the materialization of an extreme event that may occur with a 5% of probability. In economics, this idea is attractive to study the distributional effects over a macroeconomic variable. Cecchetti and Li (2008) use this method to study the impact of asset prices on the distribution of inflation and GDP growth, while De Nicolo and Lucchetta (2017) identify that this methodology provides more accurate forecasts of GDP downside risk than traditional VAR and FAVAR models. More related to financial stability the method has been used to assess the impact of systemic risk measures over the GDP growth distribution (Giglio et al., 2016; Aikman et al., 2018; Lang et al., 2019). In one of the most recent influential works, Adrian et al. (2019) also show that this methodology unmasks heterogeneous effects of financial conditions over the GDP growth distribution. Finally, more related to macroprudential policy, Duprey and Ueberfeldt (2020) use quantile regressions to assess the complementarity between macroprudential and monetary policy in Canada, and Brandao-Marques et al. (2020) also add the relationship with foreign exchange interventions by comparing the effects of these policies on the output gap distribution.

In general, quantile regression models allow for heterogeneous covariates effects to be taken into account, while the availability of panel data allow the inclusion of fixed effects to control for time-invariant unobserved heterogeneity. The application of quantile models with fixed effects is straightforward as it proceeds in a

quantile-by-quantile fashion by allowing for a different fixed effect at each quantile (Koenker, 2005). The large sample properties of these estimates are the same as standard quantile regressions when the time dimension (T) is large both in absolute terms and relative to the cross-sectional dimension (N). Certainly, the fixed effects estimator in panel quantile regressions is the equivalent to the LSDV estimator used in linear regression when T is large with respect to N (Koenker and Geiling, 2001; Kato et al., 2012).⁵

In this context, the quantile panel fixed effects model is represented as follows:

$$\hat{Q}_{y_{it}|X_{it},\alpha_i}(\tau|X_{it},\alpha_i) = \hat{\alpha}_{it} + X_{it}\hat{\beta}_\tau \quad (1)$$

$$(\hat{\beta}_\tau, \hat{\alpha}_{it}) = \arg \min_{\beta_\tau, \alpha_i} \sum_{i=1}^n \sum_{t=1}^{T-h} \rho_\tau |y_{it} - X_{it}\beta_\tau - \alpha_i| \quad (2)$$

$$\rho_\tau = \tau \cdot \mathbf{1}_{(y_{it} \geq X_{it}\beta_\tau + \alpha_i)} + (1 - \tau) \cdot \mathbf{1}_{(y_{it} < X_{it}\beta_\tau + \alpha_i)} \quad (3)$$

where, $\hat{Q}$ is the estimated quantile function; τ is a given percentile; y_{it} is the dependent variable, X_{it} is a vector of explanatory variables, α_i represents a vector of individual unobserved effects, ρ_τ are weights that depend on the quantile, and $\mathbf{1}$ is an indicator function signaling whether the estimated errors are positive or negative.

Koenker and Bassett (1978) show that $\hat{Q}_{y_{it}|X_{it}}(\tau|X_{it})$ is a consistent estimator of the quantile function of $y_{it}|X_{it}$ providing an inverse cumulative distribution function. The model is solved as an optimization problem where the weighted sum of the absolute value of the residuals is minimized, instead of the squared residuals as in linear regression.

4.2. The empirical model

When T is large in relative terms to N , including additive unobserved fixed effects into a panel quantile regression provides consistent estimates. As described further below, the large sample

⁵ However, if T is small relative to N or if T and N are of similar size, estimates of the common parameters may be biased or even under-identified, and an incidental parameters problem may arise. To solve these problems, several methods have been proposed in the literature such as penalized estimators (Koenker, 2004), subtracting the fixed effect from the variable of interest (Canay, 2011), correlated random effects (Graham and Powell, 2012), and quantile estimations via moments (Machado and Santos Silva, 2019), among others.

constructed for this study allows this approach to be pursued. Thus, the estimated baseline model is the following:

$$\hat{Q}_{y_{i,t+h}|x_{it},\alpha_i}(\tau x_{it}, \alpha_i) = \hat{\alpha}_{it} + \hat{\beta}_{1\tau} y_{it} + \hat{\beta}_{2\tau} FCI_{it} + \hat{\beta}_{3\tau} credit_{it} + \hat{\beta}_{4\tau} HP_{it} + \hat{\beta}_{5\tau} MPI_{it}; \quad (4)$$

$\tau = 5, 10, \dots, 90, 95; h = 1, \dots, 16$

where, α_i represents the unobserved country-effects; y_{it} is the contemporaneous GDP annual growth rate; FCI is the index of financial conditions; $credit$ is the 2-year average change in the credit-to-GDP ratio; HP is the 2-year average growth in house prices; MPI is an index of the use of macroprudential policy; and τ represents the 19 estimated quantiles departing from the 5th to the 95th percentiles. Finally, the variable of interest ($y_{i,t+h}$) is defined as the annualized average growth rate of real GDP for every country i at $t + h$ quarters ahead over a time horizon from 1 to 16 quarters, as follows:

$$y_{i,t+h} = \ln \left(\frac{GDP_{i,t+h}}{GDP_{i,t}} \right) / \left(\frac{h}{4} \right); h = 1, \dots, 16 \quad (5)$$

The specification in Eq. (4) extends that proposed by Adrian et al. (2019) in two main directions. First, besides the FCI, which Adrian et al. (2019) find very relevant to explain changes in the conditional GDP growth distribution at horizons between 1 and 4 quarters ahead, I add two of the most informative mid-term early warning indicators of cyclical systemic risk (see Castro et al., 2016; Lang et al., 2019; for empirical testing), which would allow GDP growth to be characterized at mid-term horizons. In this regard, Aikman et al. (2018) find that early-warning cyclical systemic risk indicators perform well when assessing mid-term GDP tail risk. Thus, since the FCI incorporates more contemporaneous information of systemic risk, it is intended to characterize the conditional GDP growth distribution at short horizons, while credit and house price growth would characterize the GDP growth distribution at longer horizons. Second, this is a panel specification, which allows more information to be used, mainly regarding the use of macroprudential policy, while accounting for country-level fixed effects.

In order to assess the goodness of fit of the models, I compute the pseudo- R^2 ($\tilde{R}^2$) proposed by Koenker and Machado (1999). This measure is dependent on the quantile, so it is a local measure of fit of the quantile specific regression and differs from the OLS R^2 . In particular, the measure compares the sum of weighted deviations for the model of interest with the same sum from a model in which only the intercept appears, and is defined as follows:

$$\tilde{R}^2(\tau) = 1 - \frac{\sum_{t=1}^T \rho_\tau(Y_{t+h} - \hat{\alpha}_i(\tau) - \mathbf{X}_t \hat{\beta}(\tau))}{\sum_{t=1}^T \rho_\tau(Y_{t+h} - \hat{\alpha}_i(\tau))} \quad (6)$$

Finally, after obtaining the estimates of the parameters in the panel quantile regressions, predictions of the variable of interest $y_{i,t+h}$ can be made for each country at every period. These predictions obtained for a given country and period of time can be computed at the different percentiles. In this application I compute it for 19 quantiles from 0.05 to 0.95 with steps of 0.05. The predicted values would shape the conditional distribution of $y_{i,t+h}$, which enable a probability density function to be estimated. However, mapping the estimates of the quantile function into a probability density function is not straightforward. There are several methods to recover the conditional density. In principle, one could obtain it by estimating a large number of quantiles or by cubic interpolation. However, these methods may produce non-monotonic conditional cumulative distribution functions. Adrian et al. (2019) propose estimating it parametrically by fitting a skewed- t distribution to

the predicted values.⁶ However, this method introduces strong assumptions regarding the density function. A non-parametric fit using kernel-based methods provides a smooth and monotone cumulative distribution function while allowing for more flexibility (Escanciano and Goh, 2014). In particular, I follow the weighted kernel interpolation method in Galvez and Mencía (2014), as described in Annex 4.

4.3. Data

Although the iMaPP database includes information on macroprudential policies in 134 countries, the final sample is smaller due to the limitations regarding the methodology and the availability of long series of quarterly data on the macrofinancial variables included in the analysis. As was described in Section 4.1, the relationship between the size of N and T is key to guarantee unbiased and asymptotic estimates in panel quantile regressions with individual effects (Kato et al., 2012). In particular, the cross-sectional dimension must be small enough with respect to the time dimension.⁷ Therefore, given that the iMaPP contains quarterly data from 1990 to 2016, a sample between 35 and 40 countries would be appropriate.

This is similar to the number of countries for which the IMF computes the FCI (24 advanced economies and 19 emerging economies). Thus, I collect data for this index from the IMF. Regarding the cyclical variables, I use the BIS as a data source. It reports quarterly data on the credit-to-GDP ratio and house prices for 43 and 58 countries, respectively. Finally, I collect quarterly data on real GDP growth from the OECD, which reports this data for 47 countries. After merging these databases, they coincide in exactly 36 countries (24 advanced economies and 12 emerging economies), which compose the final sample used for the regressions.⁸ Summary statistics of the macrofinancial variables for this sample is presented in Table 1.

Regarding macroprudential policy, I compute an index based on the one used by Alam et al. (2019) using the same database. The advantage of using an index compared to the use of dummy variables is that it allows the net macroprudential position to be accounted for by accumulating the measures over a period of time, including those taken within the same period (see Boar et al., 2017, for a discussion). The computed index consists of the 4-quarter net sum of macroprudential measures in the 17 categories defined in the iMaPP database (see Annex 1), and it is constructed as follows:

$$MPI_{it} = \sum_{j=1}^{17} \sum_{p=0}^3 SMP_{i,j,t-p}; SMP_{ijt} = \#TSMPI_{ijt} - \#LSMPI_{ijt} \quad (7)$$

where, MPI is the index for country i at quarter t , SMP encompasses the specific macroprudential measures in each category j , and, $TSMPI$ and $LSMPI$ represent the number of tightening and loosening implemented measures within each category, respectively. When a macroprudential measure in any of the defined categories is tightened during the current or the past 3 quarters, the index adds

⁶ Adrian et al. (2019) prove that fitting the probability density function using a non-parametric approach do not change the results.

⁷ Another key element is the existence of enough observations above/below the quantile to assure the fit is not an artifact of a few extreme observations. Moreover, the asymptotics of quantile regressions rely on there being enough observations on both sides in order to accomplish the conditional Central Limit Theorem. A rough rule of thumb is: $\min\{\pi\tau, n(1-\tau)\} \geq 10p$, where p is the number of explanatory variables (see, Chernozhukov, 2005).

⁸ The countries included in the final sample includes the following advanced economies: Australia, Austria, Belgium, Canada, Czech Republic, Denmark, Finland, France, Germany, Greece, Ireland, Israel, Italy, Japan, Korea, Netherlands, New Zealand, Norway, Portugal, Spain, Sweden, Switzerland, the United Kingdom, and the United States. Emerging economies: Brazil, Chile, China, Colombia, Hungary, India, Indonesia, Mexico, Poland, Russia, South Africa, and Turkey.

Table 1
Summary statistics.

Variable	p5	p25	p50	p75	p95
GDP (1y Growth, %)	-1.25	0.13	0.21	1.30	2.63
Credit-to-GDP (2y Av. Change, %)	-12.31	-1.69	4.29	10.48	22.94
House prices (2y Av. Growth, %)	-19.21	-4.52	2.86	11.75	25.93
Financial Conditions Index (FCI)	-1.26	-0.69	-0.14	0.44	1.86
Macroprudential Policy Index (MPI)	-1	0	0	1	4

Source: IMF, BIS and OECD.

1. Conversely, when a measure is loosened during this period in any of these categories the index subtracts 1. Thus, it is a 4-quarter moving measure of the net macroprudential position in each country.⁹ Similarly to other indexes constructed in the literature, the intention of this index is not to capture the intensity of the measures, which would imply knowing how binding each measure in each country is. Nonetheless, in an effort to include this type of information, the iMaPP collects data on regulatory LTV limits, which I include as an additional robustness exercise in Section 6.

This index is very similar in spirit to previous recent proposals in the literature. Cerutti et al. (2017) compute an annual index for 119 countries from 2000 to 2013, based on previous versions of IMF surveys. That index is the annual simple sum of the implemented measures in 12 categories that precede the current iMaPP classification. However, it does not include changes in the level of measures that were already in place and does not distinguish between tightening and loosening measures due to the difficulty to recognize the direction of the policy in the surveys. This distinction is crucial given that tightening and loosening policies have different characteristics as was observed in Section 2. Tightening measures are usually progressively implemented during expansionary phases of the financial cycle, while loosening measures are usually rapidly and simultaneously implemented during financial stress periods. Also, the aim of tightening measures is to increase the resilience of the financial sector in advance of potential shocks given the build-up of systemic risks; while loosening measures are aimed at mitigating the effects of the materialization of systemic risk. Boar et al. (2017) merge this database with others reported previously in Lim et al. (2011) and Kuttner and Shim (2016) to compute an index from 1990 for 64 countries.¹⁰ Nonetheless, this index also lacks the distinction between tightening and loosening measures and accounts for yearly rather than quarterly data. Kim and Mehrotra (2018) compute an index with quarterly data and that distinguishes the direction of the policy, but only for four Asian economies (Duprey and Ueberfeldt, 2020 compute a similar index for Canada). Against this background, the iMaPP provides the most complete database of macroprudential policies in terms of coverage, frequency and the ability to distinguish the direction of the policies.

5. Results

Departing from the baseline model in Eq. (4), I assess the performance of different specifications by adding one explanatory variable at a time. I start with the contemporaneous GDP growth rate, and then I add the FCI, the credit-to-GDP growth, the house

price growth, and the MPI. These regressions are estimated for GDP growth 4 and 12 quarters ahead in order to identify differences in the contribution of the included variables in characterizing the GDP growth at a short- and a mid-term horizon. Table 2 presents the pseudo-R² for each specification, computed as in Eq. (5), for the 5th, 25th, 50th, 75th, and 95th percentiles. In general, the full specification improves the goodness of fit of the model. However, the marginal contribution of the independent variables varies across quantiles and horizons. In particular, the FCI improves most of the fit of the model at the short-term horizon but it does not add value in the mid-term. This is consistent with the fact that this index captures conditions of financial stress, which usually increase when systemic risk materializes, being rapidly reflected in GDP growth. On the other hand, the early-warning variables of cyclical systemic risk are more useful in characterizing the GDP growth distribution at longer horizons. Certainly, macrofinancial variables related to credit and house prices have been previously found to capture properly the build-up of systemic risk between two and three-years before it materializes (see Lang et al., 2019). Overall, the best fit in all the cases is at the tails of the GDP growth distribution, and mainly at low quantiles. This evidences the importance of moving away from conditional mean estimations in order to model properly the dynamics of the responses of output to financial cycle movements and the effects of macroprudential policy.

In Fig. 4 I plot the estimated quantile regression coefficients in the baseline model, the 95% confidence bands obtained using bootstrapping, and a comparison against the conditional mean estimation provided by an OLS model. Results are presented for the conditional GDP growth distribution 4 and 12 quarters ahead. Two main general results are observed. First, there are clear heterogeneous effects of macrofinancial variables and macroprudential policy on the quantiles of the GDP growth conditional distribution; and, second, these effects change depending on the time horizon.¹¹

In the short-run, the impact of credit and house prices is positive on median growth but negative on the left tail of the distribution. This suggests that credit and house price developments feed economic expansions relatively fast, but it is achieved at the expense of a deterioration of growth-at-risk. At longer horizons, the effect of credit growth becomes negative at all the quantiles. Moreover, although the impact on growth-at-risk continues to be negative, its magnitude is more than three times that identified in the short-run, and almost four times larger than the impact on the median. Something similar is observed for house price growth. In this case, the effect on the median is non-significant but the impact on growth-at-risk doubles with respect to the effect at the shorter horizon.

⁹ To check the sensitiveness of the results to the index constructed in this way, I also construct two additional versions of the index, one aggregating the net measures during 8 quarters, and one accumulating them from the first to the current period. Results are found to be robust for the different models used in the study and are available upon request.

¹⁰ The database used by Lim et al. (2011) is based on an IMF survey conducted in 2010 and compiles macroprudential measures in 49 countries. Kuttner and Shim (2016) construct a database for 57 countries from 1980 focused on housing related policies.

¹¹ The estimated coefficients for contemporaneous economic growth are presented in Annex 3. At a 4-quarter horizon the estimated coefficients are significant and positive across quantiles. Nonetheless, the estimated coefficient at the 5th percentile is around four-times larger than the one in the median. This indicates that the impact of economic conditions on future growth is much more important on the downside risk than on the central tendency of GDP growth. This is similar to what is documented by Adrian et al. (2019). At a longer horizon, low positive effects are still observed in the median, while a negative coefficient is estimated on the left tail. These results suggest that in the mid-term favourable economic conditions start to represent larger downside risk.

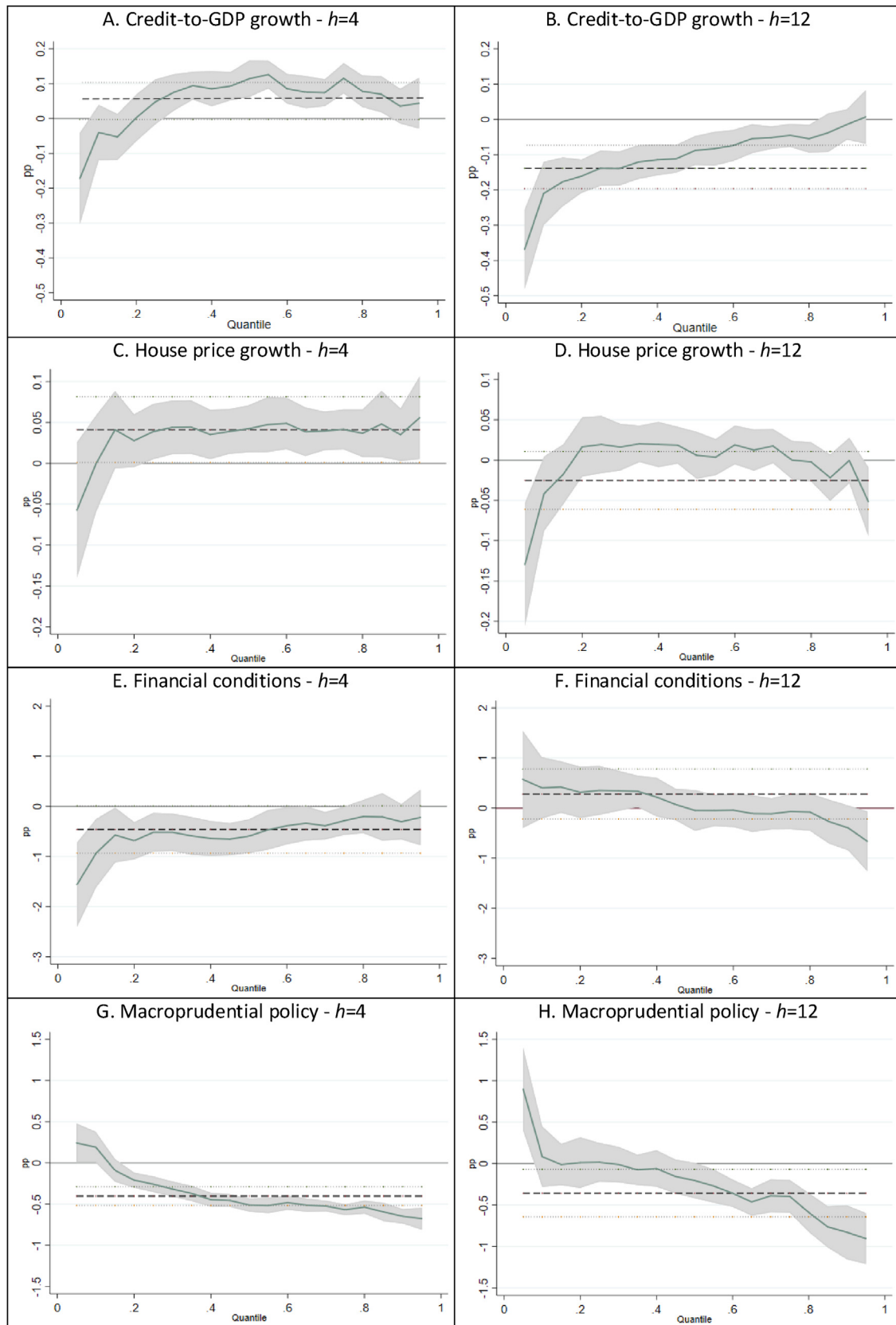


Fig. 4. Estimated quantile regression coefficients at 4- and 12-quarter horizons.

Note: The continuous lines represent the estimated coefficients at each quantile, the grey shaded areas represent the 95% confidence bands obtained using bootstrapped standard errors with 500 replications, the horizontal continuous grey line signals the value of zero, and the dark and light dashed horizontal lines represent the estimated coefficients via OLS and their 95% confidence bands, respectively.

Table 2

Performance of different specifications of quantile regressions of conditional GDP growth 4 and 12 quarters ahead at relevant quantiles.

	$h = 4$					$h = 12$				
	0.05	0.25	0.50	0.75	0.95	0.05	0.25	0.50	0.75	0.95
GDP	0.117	0.105	0.104	0.152	0.185	0.096	0.094	0.084	0.111	0.148
FCI	0.242	0.197	0.164	0.214	0.240	0.113	0.102	0.096	0.127	0.162
Credit	0.293	0.223	0.191	0.228	0.269	0.281	0.245	0.222	0.239	0.278
HP	0.315	0.237	0.209	0.236	0.272	0.364	0.296	0.273	0.293	0.325
MPI	0.369	0.278	0.242	0.275	0.324	0.418	0.342	0.314	0.342	0.386

Note: The table presents the value of the pseudo-R² obtained from quantile estimations of GDP growth 4 and 12 quarters ahead at the five quantiles in the second row. Each row represents a regression where the variable in that row is added to those in previous rows. Values in bold represent the maximum value of the pseudo-R² in each row and horizon.

This indicates that the accumulation of cyclical risk significantly deteriorates growth-at-risk in the mid-term. On the other hand, financial conditions have only significant effects on the quantiles of the GDP distribution at short horizons, when the impact is more evident in the left tail. These results are consistent with those identified by [Adrian et al. \(2019\)](#) for financial conditions and confirms that adding cyclical risk variables to the specification proposed by these authors improves the identification of growth-at-risk in the mid-term.

Regarding macroprudential policy, impact is very heterogeneous across quantiles of the GDP growth distribution. On one hand, the impact is negative on the median and the right tail of the distribution. This suggests that tightening macroprudential policy has effects on taming the business cycle, which are translated into a deterioration of median and large growth rates. This is also observed when an OLS specification is fitted and consistent with previous findings using conditional mean models to assess the impact of macroprudential policy on GDP growth. The magnitude of the impact on the median one year after the implementation (around 0.5pp) is also consistent with those studies, which have associated these negative effects to the costs of macroprudential policy ([Kim and Mehrotra, 2018](#); [Richter et al., 2019](#)). Certainly, by limiting the credit supply, macroprudential would reduce consumption and investment affecting GDP growth negatively ([Alam et al., 2019](#)).

On the other hand, tightening macroprudential policy has positive effects on growth-at-risk that are more evident in the mid-term. This confirms that benefits of macroprudential policy on GDP growth do exist but they are only observed on the left-tail of the distribution. This indicates that the increase in the resilience of the financial sector pursued by tightening macroprudential policy effectively reduces the downside risk of GDP growth in the mid-term. Moreover, the combined effect of macroprudential policy on improving growth-at-risk and slowing excessive growth rates, reduces the skewness and dispersions of the GDP growth distribution. This finding is consistent with the reduction in the output growth volatility identified by [Boar et al. \(2017\)](#) in countries actively implementing macroprudential policy. These results also suggest that it would be possible to compare the positive and negative effects on different quantiles of the GDP growth distribution. However, policymaker preferences and various other aspects that are beyond the scope of this study would need to be considered (see [Brandao-Marques et al., 2020](#), for a proposal to perform this type of analysis by computing loss functions).

5.1. The response of growth-at-risk over time

Since I identify important differences on the impact of these variables on the left-tail of the GDP growth distribution at different horizons, I extend the estimations of growth-at-risk to the entire time horizon between 1 and 16 quarters. I plot the estimated coefficients and their 95% bootstrapped confidence bands in [Fig. 5](#). This allows the evolution of the impact of these variables to be identified

over time. It is confirmed that the effect of cyclical risk and financial conditions on growth-at-risk is always negative when significant, and that financial conditions have immediate effects on growth-at-risk, while cyclical risk is more relevant at longer horizons. In particular, a 1-point increase in the FCI leads to a 4pp deterioration in growth-at-risk in the next quarter, which implies a very rapid transmission of tighter conditions. This negative effect starts to decrease until becoming non-significant after 7 quarters. This is consistent with the decreasing trend of the effect of the FCI on the GDP growth distribution identified by [Adrian et al. \(2019\)](#) between 1 and 4 quarters ahead.

Although both credit and house price growth have a significant negative impact on growth-at-risk at long horizons, they present a different term-structure. Credit growth has a significant effect on the deterioration of growth-at-risk at horizons between 4 and 15 quarters and reaches a maximum impact around 13 quarters ahead. On the other hand, the maximum impact of house prices is observed 8 quarters ahead, and the negative effect dilutes more rapidly. This suggests that an expansionary phase of the financial cycle characterized by both high credit and high house price growth would have the largest negative impact on growth-at-risk around 2-years after these vulnerabilities start to build-up.

The assessment of the response of growth-at-risk to macroprudential policy over time is interesting since the time required for macroprudential policy to work have not been fully investigated. The estimation results suggest that the impact lag on growth-at-risk is around 6 quarters. Although a positive impact can be identified earlier, it is not clearly significant. Also, the maximum benefit in reducing the downside risk of GDP growth is reached 8 quarters after the implementation of a measure and may represent around 1pp higher GDP growth. This impact would start to dilute after 13 quarters. In terms of policy implementation, this indicates that macroprudential authorities need to take early decisions on the activation or tightening of macroprudential measures in order to effectively improve growth-at-risk in the mid-term and compensate the reductions in median growth.

5.2. The impact of macroprudential policy over the financial cycle

So far, I have assumed that the effects of macroprudential policy over the GDP growth distribution remain unchanged throughout the financial cycle, which may suggest that the impact of loosening macroprudential policies would be opposite and symmetric to the impact of tightening policies. However, the use of macroprudential policy depends on the phase of the financial cycle. As I described in [Section 2](#), macroprudential instruments tend to be tightened when systemic risk builds-up during expansions of the financial cycle, while these instruments are loosened or deactivated during contractionary phases and periods of materialization of systemic risk. Moreover, the different characteristics of expansions and contractions of the financial cycle may affect the impact lag and the effectiveness of the measures. Thus, the impact on GDP growth and, particularly on growth-at-risk, may differ depending on the direc-

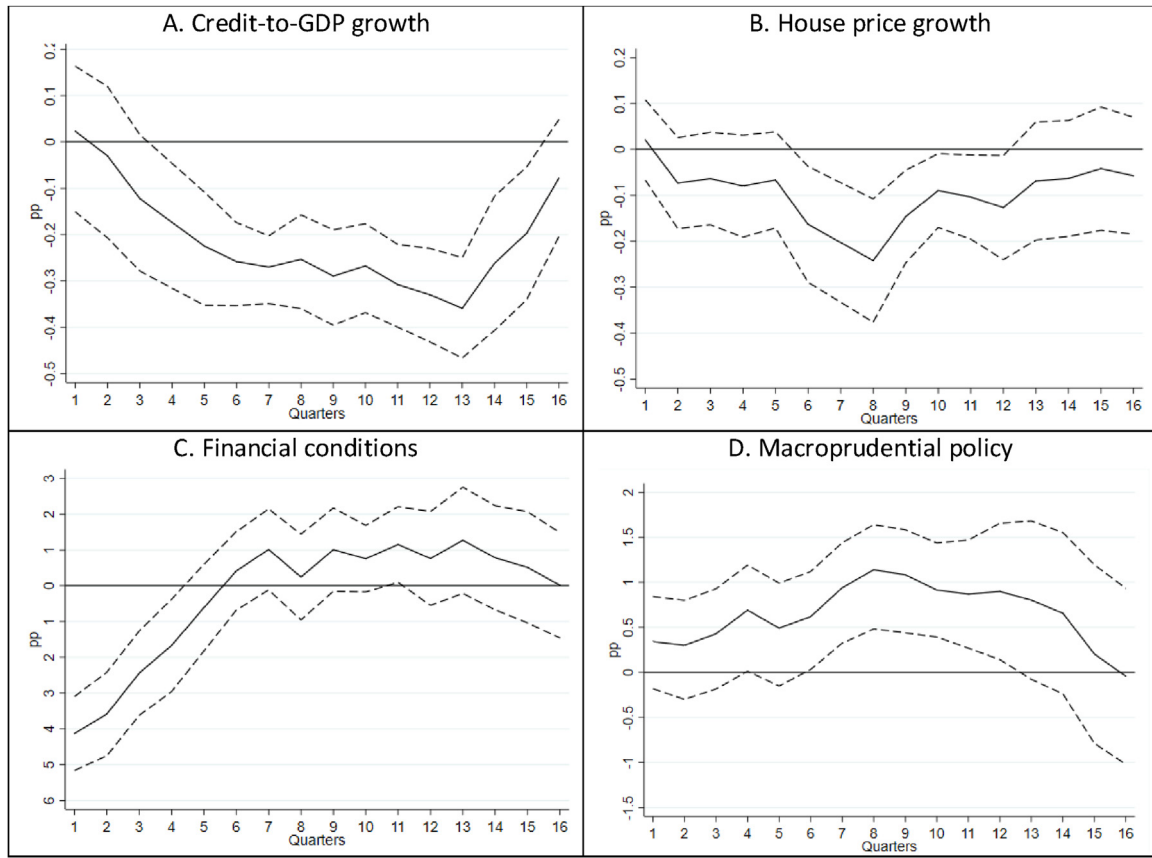


Fig. 5. Estimated coefficients of growth-at-risk regressions from 1- to 16-quarter horizons.

Note: The continuous lines represent the estimated coefficients of each variable obtained from quantile regressions on GDP growth at the 5th percentile from 1 to 16 quarters ahead. The dashed lines represent the 95% confidence bands obtained using bootstrapped standard errors with 500 replications. The horizontal lines signal the value of zero. The vertical axes represent the marginal effect on growth-at-risk in terms of percentage points.

tion of the policy, the position in the financial cycle and the time elapsed since these measures were implemented.

Certainly, previous studies have found that the timing of macroprudential policies is relevant and may produce different effects depending on the position in the financial cycle. Jiménez et al. (2016) find that accumulating capital buffers during credit expansions reduces the magnitude of bank credit contractions during systemic crises, but that it magnifies the negative effect on credit and firm's probability of survival if they are accumulated during a downward phase. Also, in an analysis of borrower-based measures, Claessens et al. (2013) and Cerutti et al. (2017) identify that these measures are more effective in booms than in bust periods.

Therefore, in order to account for different phases of the financial cycle and their interaction with macroprudential policy, I include two additional variables that identify booming and financial crises periods, and interact them with the MPI. The specification takes the following form:

$$\hat{Q}_{y_{i,t+h}|x_{it},\alpha_i}(\tau x_{it}, \alpha_i) = \hat{\alpha}_{it} + \hat{\beta}_{1\tau} y_{it} + \hat{\beta}_{2\tau} FCI_{it} + \hat{\beta}_{3\tau} credit_{it} + \hat{\beta}_{4\tau} HP_{it} \\ + \hat{\beta}_{5\tau} MPI_{it} + \hat{\beta}_{6\tau} Boom_{it} + \hat{\beta}_{7\tau} Crisis_{it} + \hat{\beta}_{8\tau} MPI_{it} Boom_{it} \\ + \hat{\beta}_{9\tau} MPI_{it} Crisis_{it}, \quad (8)$$

where $Boom_{it}$ and $Crisis_{it}$ are two binary variables that take the value of 1 if a country i experiences either of the two situations at time t . In particular, booming periods are classified as those where both credit and house prices present a 2-year average annual growth greater than 5% and a negative FCI that signals loose financial conditions. The combination of these three characteristics has

been observed during previous periods of large build-ups of cyclical systemic risk, that have also incentivized a more active use of tightening macroprudential measures (see Fig. 2). On the other hand, crisis periods are those identified as financial crises in Laeven and Valencia (2018).

In Fig. 6 I plot the response of growth-at-risk over time to the tightening of one macroprudential policy measure during booming periods and the loosening of one measure during financial crisis periods. Although both tightening policies in expansions and loosening policies in financial crises have a positive impact on growth-at-risk, there are important differences in the impact lag and the persistence of their effects. In particular, tightening macroprudential policy during expansionary phases of the financial cycle takes around 6 quarters on evidencing a significant impact on growth-at-risk and the largest economic impact (around 1.4pp) is reached 10 quarters after the implementation. Nonetheless, the positive effect is quite persistent, only starting to dilute almost 16 quarters after the policy implementation.

In contrast, the benefits of loosening macroprudential policy during financial crises are rapidly materialized but they are less persistent. A significant positive impact is observed as early as 3 quarters after one measure is loosened and the maximum benefit (around 1pp) is reached just 4 quarters afterwards. Nonetheless, the benefits become non-significant 7 quarters after loosening a macroprudential measure.

These results suggest that during large expansions of the financial cycle, characterized by loose financial conditions and buoyant credit stimulated by the feedback loop from house prices, the effect of tighter macroprudential policy on increasing resilience presents

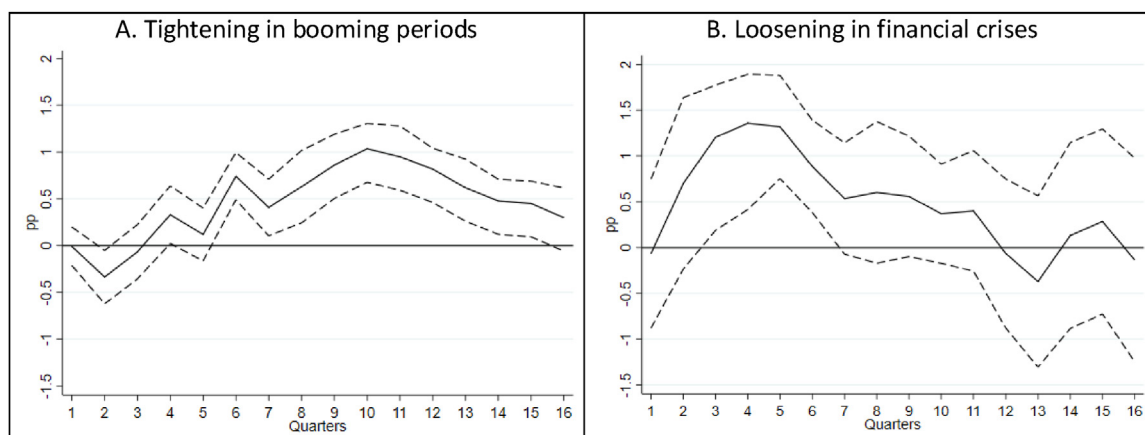


Fig. 6. Response of growth-at-risk to changes in macroprudential policy by direction of the policy and position in the financial cycle.

Note: The continuous lines represent the estimated coefficients of each variable obtained from quantile regressions on GDP growth at the 5th percentile from 1 to 16 quarters ahead. The dashed lines represent the 95% confidence bands obtained using bootstrapped standard errors with 500 replications. The horizontal lines signal the value of zero. The vertical axes represent the marginal effect on growth-at-risk in terms of percentage points.

an important delay. The difference in the economic impact of each type of policy under these scenarios can also be related to the fact that, during financial expansions, tightening decisions are usually taken gradually, to the extent that signals of the build-up of systemic vulnerabilities are confirmed; thereby, tightening measures are more distributed along the expansionary phase of the cycle. On the contrary, when macroprudential policy is loosened, it is usually a fast process implemented in few steps. That is because financial crises materialize quickly forcing macroprudential authorities to employ loose measures rapidly. Also, loosening macroprudential policy promptly would avoid excessive deleveraging under an adverse scenario, and the benefits in the resilience of the financial sector could be more rapidly observed.

The heterogeneous effects of both macrofinancial conditions and macroprudential policy across quantiles of the GDP growth distribution would affect its shape. In order to illustrate this effect, I map the estimates of the quantile functions obtained with the model in Eq. (8) into a probability density function, by using the kernel-based method described in Section 4.2. In particular, I show how the shape of the GDP growth distribution changes when moving from a normal times scenario (baseline case) to a large expansionary phase of the financial cycle classified as a booming period, and to a situation of financial crisis. Departing from these scenarios, I estimate the impact of implementing one macroprudential measure in the expected direction on the GDP growth distributions. I select the horizons according to the maximum impact of macroprudential policy on growth-at-risk identified above under each situation (i.e. 4 and 10 quarters ahead).

In the left panel of Fig. 7, I plot the change in the 10-quarters-ahead GDP growth distribution when moving from an average normal times situation (black line) to a large expansionary phase of the financial cycle (red line). In general, the distribution moves to the right but becomes highly left-skewed. That is, while central tendency measures signal a higher expected growth rate and the right tail seems to move proportionally, the left-tail becomes very fat indicating a high increase in the downside risk of GDP growth. This confirms the asymmetric impact of the financial cycle on different quantiles of the GDP growth distribution, and that the higher expected growth promoted by a large expansion of the financial cycle arrives at the expense of higher downside risk. The role of macroprudential policy under this scenario becomes evident when the impact of tightening macroprudential policy is mapped into a density function (blue line). The main result is that it reduces the skewness of the distribution and makes it more symmetric. In particular, both tails of the distribution shrank indicating an

important reduction of risk, while the negative impact on central tendency measures of growth seems lower. The improvement on growth-at-risk is evident, although only one measure would not be enough to compensate the deterioration on downside risk. These results are consistent with those identified by Brandao-Marques et al. (2020) on the output gap distribution when studying the complementarity between macroprudential and monetary policy after changes in financial conditions. Nonetheless, the authors only account for changes in the FCI, which do not adequately characterize GDP growth in the mid-term and do not capture properly the build-up of cyclical vulnerabilities.

On the other hand, in the right panel I show the change in the shape of the 4-quarters-ahead GDP growth distribution when moving from a normal times situation (black line) to a financial crisis scenario (red line) and the effect of loosening macroprudential policy (blue line). In particular, when entering into a financial crisis, the distribution moves to the left and shrinks the right tail; but the left-tail expands, by worsening growth-at-risk more than proportionally with respect to central tendency measures. In such a scenario, loosening macroprudential policy contracts the left-tail of the distribution, thereby improving growth-at-risk. The positive impact on median growth is less evident, while the right-tail is almost unaltered. This leads to a more symmetric and less disperse GDP growth distribution.

Overall, tightening macroprudential policy in expansions and loosening macroprudential policy in financial crises reduce the skewness and the asymmetry of the future GDP growth distribution. This is mainly achieved by shrinking the left-tail of the distribution, where the benefits of macroprudential policy concentrates; thereby reducing the downside risk of GDP growth.

5.3. Capital and borrower-based measures

Above, I disentangled the large heterogeneous effects of macroprudential policy on the GDP growth distribution depending on its direction, the position in the financial cycle and the time elapsed from its implementation. However, the definition of macroprudential policy used is an aggregation of very different types of instruments that may differ in their effectiveness and term structure. While capital measures raise the cost of funding, thereby discouraging the granting of new credit; borrower-based measures impose lending standards, thereby limiting risk-taking.

Certainly, borrower-based and capital measures are among the most used types of macroprudential instruments both in advanced and emerging economies (see Annex 2). This is why most pre-

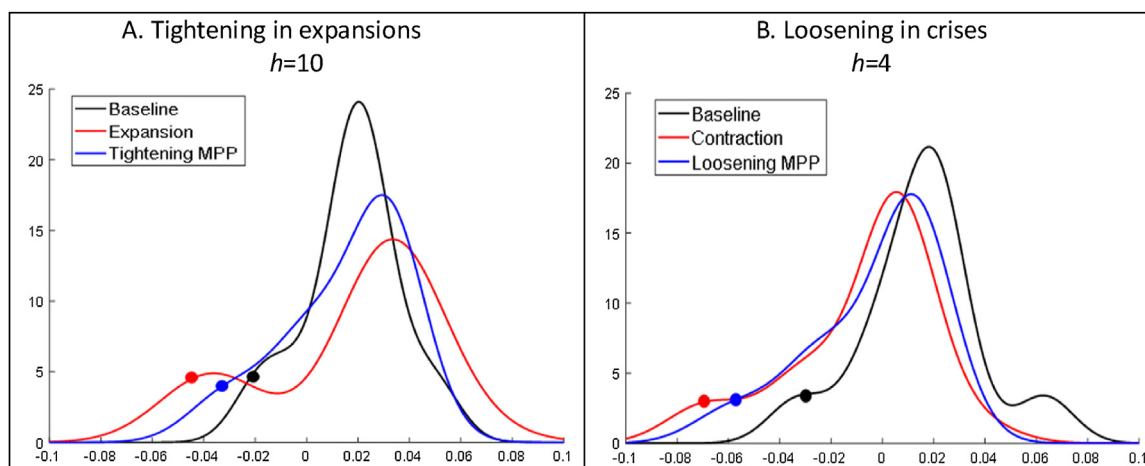


Fig. 7. Conditional GDP growth distribution 4 and 10 quarters after the implementation of tightening/loosening macroprudential policies over the cycle.

Note: The figures present the estimated GDP growth distributions at the specified horizons constructed by mapping the fitted values of the quantile functions obtained with the model in Eq. (8) into a probability density function using the kernel-based method described in Annex 4. The black densities represent average values under a normal times situation, defined as periods not classified as booms nor crises; the red densities represent the average values in a situation of financial crises (left panel), and an expansion of the financial cycle (right panel), following the classifications described above. The blue densities represent the GDP growth distribution after tightening (left panel) or loosening (right panel) one macroprudential measure departing from the crises and boom scenarios.

vious studies assessing the impact of macroprudential policies have focused on these two types of measures. Studies on capital measures have evidenced important differences in the effects of these instruments over the cycle (see Jiménez et al., 2017). In general, these effects have been identified as being negative on credit growth during upswings but positive during downturns. Regarding borrower-based measures, these instruments have been found to effectively curb house prices and households credit growth. These measures have also been identified to have different effects over the cycle. Claessens et al. (2013) and Cerutti et al. (2017) find that these measures are more effective during expansionary phases of the cycle. In this regard, Poghosyan (2019) identifies that tightening borrower-based measures during booming periods takes more time to become effective compared to loosening them during busts.

To study the potential different effects of these two types of instruments, I split the MPI into three indexes. The first one encompasses the three categories related to capital-based measures (i.e. capital requirements, capital conservation buffers and countercyclical capital buffers). The second one includes two categories regarding limits on the LTV and DSTI/LTI ratios. And, the third one aggregates the remaining 12 categories, which would control for the effect of other types of measures placed simultaneously.¹² These indexes are constructed following the same formula in Eq. (2) by aggregating the corresponding categories. All three indexes are interacted with the boom and financial crises dummy variables in order to account for the effects of the cycle. The regressions are estimated for the entire GDP growth horizon from 1 to 16 quarters ahead.

In Fig. 8 I plot the response of growth-at-risk and median GDP growth to tightening and loosening these types of policies over different horizons by accounting for the position in the financial cycle. Consistent with my previous findings using the MPI, I identify a positive impact of capital measures on the downside risk of GDP growth both when tightening during large expansionary phases and loosening during financial crises. Similarly, the impact lag differs importantly under these two scenarios. On one hand, the benefits of tightening capital measures in expansions takes more than 6 quarters to become evident and between 2 and 3 years to reach their maximum economic impact (around 1pp). On the other

hand, when these measures are loosened during financial crises, the benefits are immediate and larger. In particular, the maximum benefit could reach more than 1.5pp just 5 quarters after the implementation.

Compared to growth-at-risk, the effects on median growth are lower but the sign changes depending on the direction of the policy in each scenario. In expansions, tightening capital measures would reduce median growth by up to 0.5pp, which can be related to the effect on smoothing the cycle. Certainly, this is a secondary objective of countercyclical capital buffers. Conversely, during financial crises, releasing buffers would promote a stronger and more persistent economic recovery. These results are consistent with previous literature using conditional mean models. Noss and Toffano (2016) identify that increasing the banks' capital ratios leads to a reduction of GDP growth in the mid-term. Jiménez et al. (2017) also identify that a 1pp increase in capital buffers leads to a 4pp reduction on credit growth over expansionary phases, while releasing the same amount of capital during busts implies a 9pp lower reduction in bank credit to firms during the following 12 quarters. The authors identify similar non-linearities in terms of the probability of survival of firms and employment.

Similar to capital instruments, tightening borrower-based measures during large expansions of the financial cycle has important benefits on growth-at-risk that seem to be larger than the negative impact on median growth. However, in contrast to capital measures, the benefits of limiting LTV or DSTI/LTI ratios are rapidly materialized and would persist for more than 2 years after their implementation. On the other hand, the impact of loosening borrower-based measures during financial crises is more limited, both on the left tail and the median of the GDP growth distribution. The positive impact on growth-at-risk would be clearly significant only between 4 and 5 quarters after loosening these instruments, and the magnitude of the benefit is lower than that of loosening capital measures. These results are consistent with previous findings on the effectiveness of borrower-based measures. Although previous approaches estimate conditional mean effects, those studies have found larger effects when these measures are tightened during expansions than when they are loosened during contractions. Claessens et al. (2013) and Cerutti et al. (2017) document this heterogeneous impact on credit and house price growth, while Galán and Lamas (2019) identify these differences in terms of default risk of mortgages.

¹² The definition of the categories is presented in Annex 1.

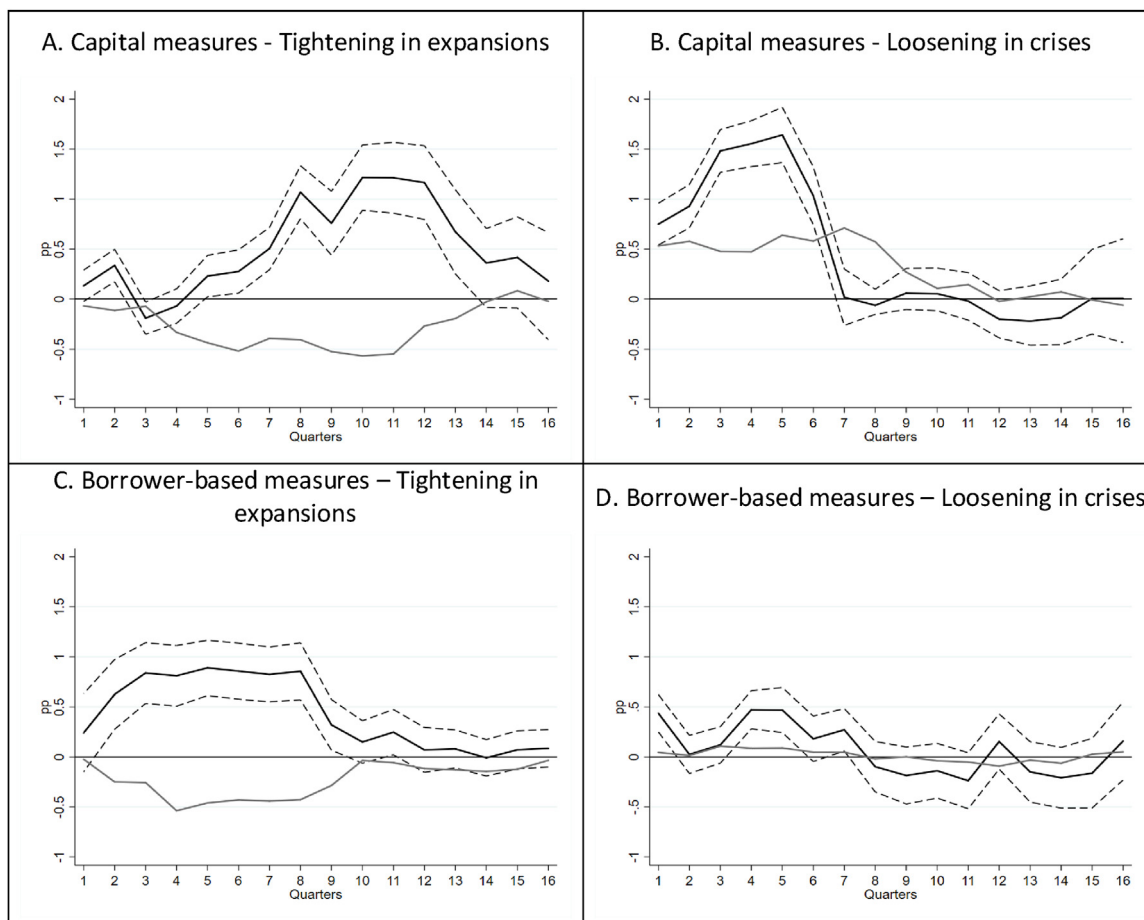


Fig. 8. Response of growth-at-risk and median growth to tightening and loosening capital and borrower-based measures over time.

Note: The dark black and light grey continuous lines represent the estimated coefficients of each variable obtained from quantile regressions on GDP growth at the 5th and the 50th percentiles from 1 to 16 quarters ahead, respectively. The dashed lines represent the 95% confidence bands of the estimated coefficients at the 5th percentile obtained using bootstrapped standard errors with 500 replications. The horizontal lines signal the value of zero. The vertical axes represent the marginal effect on growth-at-risk in terms of percentage points.

Overall, these results have important policy implications. Although, both types of measures are effective in reducing the downside risk of GDP growth during expansionary phases of the cycle, the identified impact delay suggests that capital measures should be implemented promptly and earlier than borrower-based measures, whose benefits can be perceived more rapidly. During downturns, the benefits of loosening capital requirements or releasing capital buffers on reducing the downside risk of GDP growth are important and more immediate, while the benefits of loosening borrower-based measures are more limited. Certainly, given that risks materialize rapidly during crises, macroprudential authorities have incentives to release the entire range of capital buffers in order to rapidly improve banking sector resilience. On the other hand, it is very likely that loosening borrower-based measures during busts would not be binding given that the unfavourable macrofinancial environment incentivises banks to tighten credit conditions.

6. Robustness

6.1. A measure of intensity: mean regulatory LTV

Above, I identified the impact of macroprudential policy using narrative measures that distinguish the direction of the implemented policy and the type of instrument. However, I have omitted the magnitude of the measures given the complexity of creating

intensity measures for the different types of instruments in the sample. Nonetheless, in the case of limits on the LTV ratio, the iMaPP database provides information on the regulatory LTV for a wide sample of countries. I take advantage of this information to assess the effects of this measure on growth-at-risk. Certainly, most of the previous studies assessing the effectiveness of macroprudential measures have focused on LTV limits given the availability of information. [Claessens et al. \(2013\)](#) find that LTV caps reduce asset prices and leverage growth. [Akinci and Olmstead-Rumsey \(2018\)](#) identify that limits on LTV ratios are effective in curbing credit growth and house price appreciation. [Alam et al. \(2019\)](#) also finds that tightening LTV limits has important effects on reducing credit growth but small effects on dampening consumption in the short-run. In terms of GDP growth, [Richter et al. \(2019\)](#) find that this instrument has negative effects on output growth in the mid-term. Following the results obtained above, I would expect to observe benefits of LTV limits on growth-at-risk. In this context, this exercise would not only allow the robustness of the previous results obtained to be checked using narrative measures of macroprudential policy, but also it enables the estimation of the elasticity to specific changes in LTV limits in different phases of the financial cycle. Thus, I estimate the panel quantile model in Eq. (8) after replacing the MPI by the mean regulatory LTV.

In [Fig. 9](#), I plot the estimated responses of growth-at-risk and median growth to a 10pp change in the LTV limit over a 16-quarter horizon. As before, I distinguish between tightening during expan-

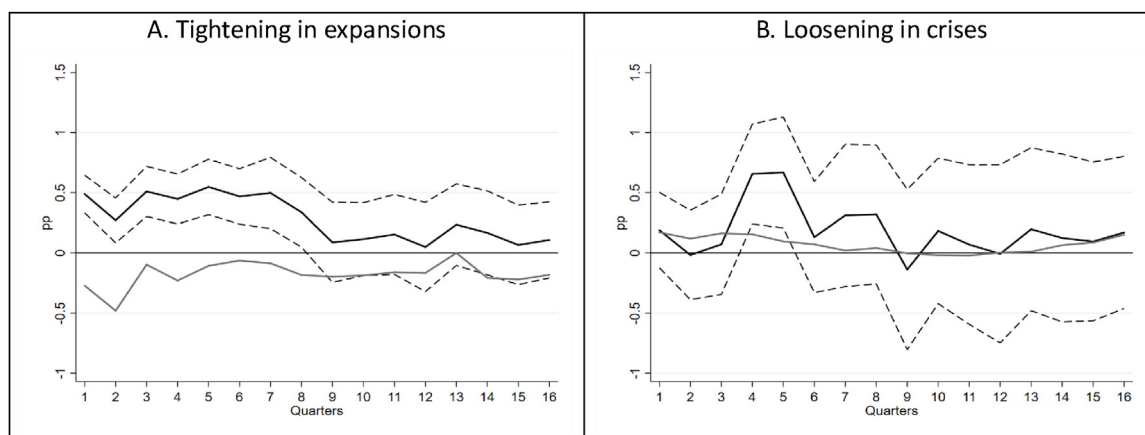


Fig. 9. Response of growth-at-risk and median growth to a 10pp change in the LTV cap.

Note: The dark black and light grey continuous lines represent the estimated coefficients of each variable obtained from quantile regressions on GDP growth at the 5th and the 50th percentiles from 1 to 16 quarters ahead, respectively. The dashed lines represent the 95% confidence bands of the estimated coefficients at the 5th percentile obtained using bootstrapped standard errors with 500 replications. The horizontal lines signal the value of zero. The vertical axes represent the marginal effect on growth-at-risk in terms of percentage points.

sions of the financial cycle and loosening during financial crises. In the former scenario, a 10pp decrease in the LTV limit would improve growth-at-risk by around 0.5pp. This positive impact is immediately observed and would persist for 8 quarters. On the other hand, median growth would be negatively affected by this policy. Although this decrease could represent a reduction of around 0.4pp in GDP growth in the following 2 quarters, this negative impact is low persistent. On the other hand, softening the LTV limit in 10pp during a systemic event leads to benefits both in the left-tail and the median of the GDP growth distribution. However, the positive impact on growth-at-risk is only significant for a couple of periods around one-year after the implementation. These results are fully consistent with those obtained in Section 5.3 using a narrative measure of borrower-based policies

6.2. Emerging and advanced economies

Although macroprudential policy has been widely used both in advanced and emerging economies, the preferred type of instruments differs. While advanced economies are very active in the use of LTV limits, the most common instrument used by emerging economies are limits to foreign exchange positions (see Annex 2). Also, emerging economies have been more active in the use of macroprudential instruments during the 90's and the years before the last global financial crisis. This heterogeneity may affect the impact of macroprudential policy on the GDP growth distribution. To explore this, I split the sample into 24 advanced and 12 emerging economies following the IMF classification and re-estimate the model in Eq. (8) for the 5th and the 50th percentiles over the complete GDP growth horizon from 1 to 16 quarters.

In Fig. 10 I present the growth-at-risk and the median growth responses to tightening and loosening one macroprudential measure during financial expansions and crises, respectively. In general, results are consistent with those obtained above with the complete sample. Tightening macroprudential policy during expansions of the financial cycle lead to an improvement of growth-at-risk, which is only significant in the mid-term; while the benefits of loosening during financial crises are more rapidly observed. Nonetheless, some differences are observed between the two types of countries. In particular, the benefits of both tightening and loosening macroprudential policy on growth-at-risk are observed earlier in advanced economies, and especially when macroprudential policy is eased. Certainly, in emerging economies the benefits of

macroprudential policy tend to present a longer impact lag. In the case of tightening macroprudential policy during expansions, the improvement in growth-at-risk is only statistically significant 8 quarters after its implementation. Similarly, when macroprudential policy is eased during systemic events the benefits present an impact lag of around 4 quarters. These differences can be related to the preferred type of instruments used in emerging economies such as reserve requirements and limits to foreign exchange positions. There is also an important difference in the width of the confidence bands of the estimates between both samples. The estimates for emerging economies present more uncertainty, which might be related to a larger heterogeneity of the impact of macroprudential policy across countries within this sample.

7. Conclusions

Although most previous studies have identified benefits of macroprudential policy in curbing credit and house price growth (Claessens et al., 2013; Cerutti et al., 2017), and reducing the probability of systemic crises and bank's defaults (Dell'Ariccia et al., 2016; Altunbas et al., 2018); in general, the impact of macroprudential policy on GDP growth has been identified to be negative (Kim and Mehrotra, 2018; Richter et al., 2019). By extending the recent proposal by Adrian et al. (2019) on the use of quantile regressions of GDP growth, I confirm that this is the result of using methods that focus on the conditional mean. Certainly, I identify large heterogeneous effects of macroprudential policy on the quantiles of the GDP growth distribution. In particular, macroprudential policy affects the shape of the GDP growth distribution by reducing its left-skewness and offsetting the opposite effect produced by the increase of cyclical risk and the materialization of systemic events.

The method allows me to uncover important benefits of macroprudential policy on the left-tail of the GDP growth distribution, which contrast with the negative effects identified in the median and the right tail. I also find that the statistical and economic significance of these benefits is highly dependent on the position on the financial cycle, the direction of the policy, the time elapsed from the implementation, and the type of instrument used. The direction of the policy and the position in the financial cycle have a relevant role in determining the magnitude and speed of the effects of macroprudential policy on growth-at-risk. In particular, tightening macroprudential policy during expansionary phases of the

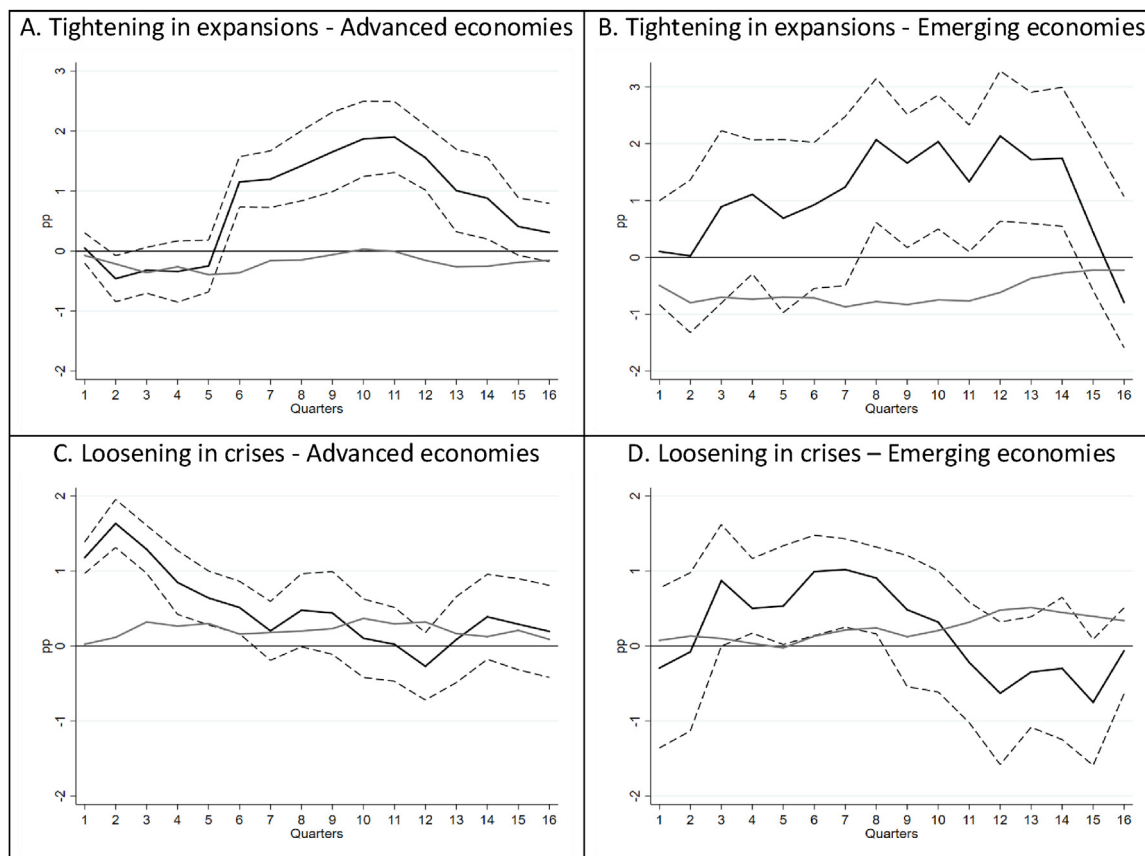


Fig. 10. Response of growth-at-risk and median growth to tightening and loosening macroprudential policy over the cycle in advanced and emerging economies. Note: The dark black and light grey continuous lines represent the estimated coefficients of each variable obtained from quantile regressions on GDP growth at the 5th and the 50th percentiles from 1 to 16 quarters ahead, respectively. The dashed lines represent the 95% confidence bands of the estimated coefficients at the 5th percentile obtained using bootstrapped standard errors with 500 replications. The horizontal lines signal the value of zero. The vertical axes represent the marginal effect on growth-at-risk in terms of percentage points.

financial cycle has large benefits in the mid-term, while loosening measures have a more immediate positive effect on reducing the downside risk of GDP growth during crises. These findings suggest the importance of timing in macroprudential policy.

I also identify important differences depending on the type of macroprudential instrument implemented. While the benefits of borrower-based measures on growth-at-risk are manifested very rapidly and tend to be persistent, the positive effects of capital measures present a lag of around 8 quarters. These results have important policy implications. Although I find that both types of measures are effective in reducing the downside risk of GDP growth, capital measures should be implemented early enough in the cycle, while borrower-based measures could also be tightened in advanced stages given that their benefits are perceived more rapidly. Conversely, during crises the benefits of releasing capital buffers are more immediate, while those of loosening borrower-based measures are very limited. Certainly, releasing capital buffers during busts allows banks to increase their resilience immediately, while softening caps on lending standards may not have real effects given that banks have incentives to tighten their credit conditions due to the unfavourable macrofinancial environment.

Overall, this study uncovers the benefits of macroprudential policy on growth-at-risk, which may offset the deterioration produced by the build-up of cyclical vulnerabilities and the materialization of financial crises. Moreover, the presented framework has useful implications for the timing of macroprudential policy decisions over the cycle and for a complete assessment of these policies in terms of GDP growth.

Appendix A. Definition of the macroprudential policy categories

Category	Definition
Capital Conservation Buffer (CCoB)	Requirements to maintain a capital buffer that is fixed over time.
Countercyclical Capital Buffers (CCyB)	Requirements to maintain capital buffers that vary over time.
Capital requirements	Other capital requirements including minimum capital requirements, risk weights and systemic risk buffers.
Leverage limits	Limits on the leverage of banks (ratios of a measure of capital over non-risk-weighted exposures).
Limits on credit growth	Limits on the growth rate or the volume of aggregate credit or sectoral credit, including penalties for credit growth above certain thresholds.
Limits on foreign currency loans	Limits and rules on loans denominated in foreign currency.
Limits on foreign exchange positions	Limits on net or gross open foreign exchange positions, exposures and funding, as well as regulation on currency mismatches.
Limits on the loan-to-deposits ratio	Limits to the loan-to-deposit ratio, including penalties for ratios above certain thresholds.
Limits on Loan-to-Value (LTV) ratio	Limits to the loan-to-value ratios targeted at housing loans, automobile loans, and commercial real estate loans.
Limits on the Debt-Service-to-Income (DSTI) and Loan-to-Income (LTI) ratios	Limits to the debt-service-to-income ratio and the loan-to-income ratio targeted at housing loans, consumer loans, and commercial real estate loans.

Loan restrictions	Strict loan limits and prohibitions, which may be conditioned on loan characteristics, bank characteristics or other factors.
Loan loss provisions	Loan loss provision requirements for macroprudential purposes, including dynamic provisioning and sectoral provisions.
Liquidity requirements	Minimum requirements for liquidity coverage ratios, liquid asset ratios, net stable funding ratios, core funding ratios and external debt restrictions that do not distinguish currencies.
Reserve requirements	Reserve requirements identified as those used for macroprudential purposes.
Systemically Important Financial Institutions (SIFIs) surcharges	Measures taken to mitigate risks from global and domestic systemically important financial institutions, including capital and liquidity surcharges.
Taxes and levies	Taxes and levies applied to specified transactions, assets, or liabilities.
Other measures	Other macroprudential measures not captured in the above categories, such as stress testing, restrictions on profit distribution, and structural measures.

Source: IMF's iMaPP database and [Alam et al. \(2019\)](#).

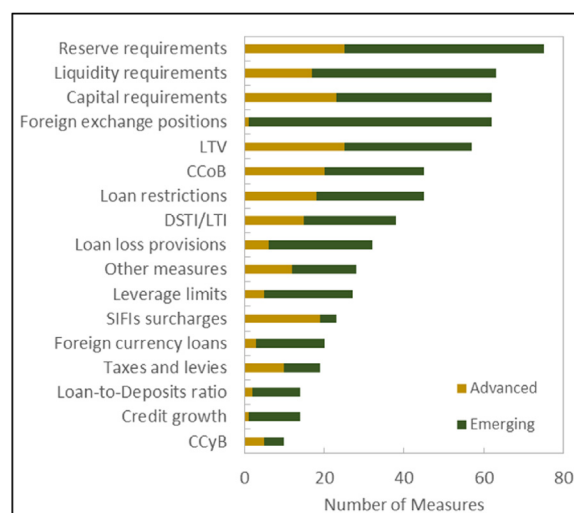


Fig. B1. Number of implemented macroprudential policies by category and type of country (1990-2016).

Source: IMF's iMaPP database.

Appendix B. The use of macroprudential policy

Figs. B1 and B2.

Appendix C. Contemporaneous GDP growth coefficients

Fig. C1

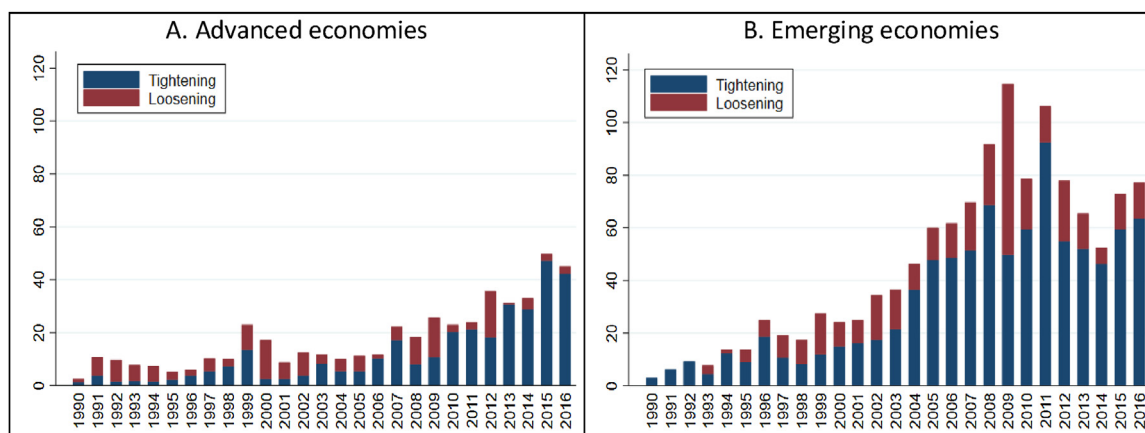


Fig. B2. Evolution of the implementation of macroprudential policy by its direction and type of country.

Source: IMF's iMaPP database. Own elaboration.

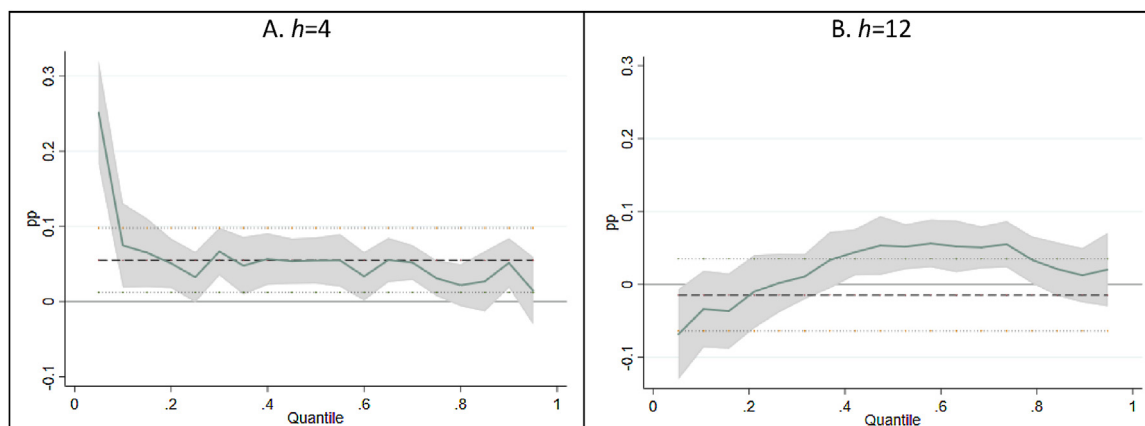


Fig. C1. Estimated quantile regression coefficients of contemporaneous GDP growth.

Note: The vertical axes represent the values of the coefficients, the horizontal axes represent the quantiles of the GDP growth distribution, the continuous dark green line represents the value of the estimated coefficients at every quantile, the grey shaded areas represent the 95% confidence bands obtained using bootstrapped standard errors with 500 replications, the horizontal red line represents the value of zero, and the dark and light dashed horizontal lines represent the estimated coefficients via OLS and their corresponding 95% confidence bands.

Appendix D. Mapping the quantiles estimates into a Kernel-based density

A parametric fitting although practical, introduces strong assumptions on the density function. A non-parametric fit using Kernel-based methods provides a smooth and monotone cumulative distribution function (*cdf*) while allowing for more flexibility (Escanciano and Goh, 2014). In particular, I follow the weighted Kernel interpolation method in Galvez and Mencía (2014), where the Kernel *cdf* would be represented by:

$$\sum_{j=1}^p w_j \Phi \left(\frac{x - q(\theta_j)}{B} \right)$$

where $\Phi(\cdot)$ is the standard Gaussian *cdf*; p is the number of points, which in this application is equal to 19 (0.05, 0.10, ..., 0.95); θ_j represents the quantile j ; B is the smoothing parameter; and w_j represents the weights ($w_1, w_2, \dots, w_p$) that minimize the squared distance between the quantile level and its associated *cdf*.

The bandwidth is computed as $B = 1.06 \min(\hat{s}, \hat{r}) p^{-1/5}$, where $\hat{s}$ is the standard deviation and $\hat{r}$ is the interquartile range of the quantile functions.

After differentiating the Kernel *cdf*, the following conditional density is obtained:

$$\frac{1}{B} \sum_{j=1}^p \hat{w}_j \phi \left(\frac{x - q(\theta_j)}{B} \right)$$

where $\phi(\cdot)$ is the standard normal density function.

References

- Adrian, T., Boyarchenko, N., Giannone, D., 2019. Vulnerable growth. *Am. Econ. Rev.* 109 (4), 1263–1289.
- Aikman, D., Haldane, A.G., Nelson, B.D., 2015. Curving the credit cycle. *Econ. J.* 125, 1072–1109.
- Aikman, D., Bridges, J., Burgess, S., Galletly, R., Levina, I., O'Neill, C., Varadi, A., 2018. Measuring Risks to Financial Stability. Staff Working Paper 738. Bank of England.
- Aikman, D., Bridges, J., Kashyap, A., Siebert, C., 2019a. Would Macroprudential Regulation Have Prevented the Last Crisis? *J. Econ. Perspect.* 33 (1), 107–130.
- Aikman, D., Bridges, J., Hacıoğlu, H., S. O'Neill, C., Raja, A., 2019b. Credit, Capital and Crises: a GDP-at-risk Approach. Staff Working Paper 724. Bank of England.
- Akinci, O., Olmstead-Rumsey, J., 2018. How effective are macroprudential policies? An empirical investigation. *J. Financ. Intermediation* 33 (C), 33–57.
- Alam, Z., Alter, A., Eiseman, J., Gelos, G., Kang, H., Narita, M., Nier, E., Wang, N., 2019. Digging Deeper—Evidence on the Effects of Macroprudential Policies From a New Database. IMF Working Paper WP/19/66. International Monetary Fund.
- Altunbas, Y., Binici, M., Gambacorta, L., 2018. Macroprudential policy and bank risk. *J. Int. Money Finance* 81, 203–220.
- Bedayo, M., Estrada, A., Saurina, J., 2020. Bank capital, lending booms, and busts. Evidence from Spain in the last 150 years. *Lat. Am. J. Central Bank.* 1, 100003.
- Boar, C., Gambacorta, L., Lombardo, G., Pereira da Silva, L., 2017. What Are the Effects of Macroprudential Policies on Macroeconomic Performance? BIS Quarterly Review, pp. 71–88, September.
- Brandao-Marques, L., Gelos, G., Narita, M., Nier, E., 2020. Leaning Against the Wind: A Cost-Benefit Analysis for an Integrated Policy Framework. IMF Working Paper WP/20/123. International Monetary Fund.
- Canay, I.A., 2011. A simple approach to quantile regression for panel data. *Econom.* 14 (3), 368–386.
- Castro, C., Estrada, A., y Martínez, J., 2016. The Countercyclical Capital Buffer in Spain: an Analysis of Key Guiding Indicators. Working Papers No. 1601. Bank of Spain.
- Cecchetti, S., Li, H., 2008. Measuring the Impact of Asset Price Booms Using Quantile Vector Autoregressions. Working Paper. Department of Economics, Brandeis University, USA.
- Cerutti, E., Claessens, S., Laeven, L., 2017. The use and effectiveness of macroprudential policies: new evidence. *J. Financ. Stab.* 28, 203–224.
- Chernozhukov, V., 2005. Extremal quantile regression. *Ann. Stat.* 33 (2), 806–839.
- Claessens, S., Kose, M.A., Terrones, M., 2012. How do business and financial cycles interact? *J. Int. Econ.* 87 (2012), 178–190.
- Claessens, S., Ghosh, S., Mihet, R., 2013. Macro-prudential policies to mitigate financial system vulnerabilities. *J. Int. Money Finance* 39, 153–185.
- De Nicolò, G., Lucchetta, M., 2017. Forecasting tail risks. *J. Appl. Econom.* 32, 159–170.
- Dell'Ariccia, G., Igan, D., Laeven, L., 2016. Credit booms and macro-financial stability. *Econ. Policy* 31, 299–355.
- Drehmann, M., Borio, C., Gambacorta, L., Jimenez, G., Trucharte, C., 2010. Countercyclical Capital Buffers: Exploring Options. BIS Working Papers No 317. Bank for International Settlements.
- Duprey, T., Ueberfeldt, A., 2020. Managing GDP Tail Risk. Staff Working Paper 202/03. Bank of Canada.
- Escanciano, J.C., Goh, C., 2014. Specification analysis of linear quantile models. *J. Econom.* 178 (3), 495–507.
- European Central Bank (ECB), 2009. Financial Stability Review. Frankfurt Am Main, November.
- European Systemic Risk Board (ESRB), 2015. Annual Report 2014. Frankfurt Am Main, July.
- Financial Stability Board (FSB), International Monetary Fund (IMF), Bank of International Settlements (BIS), 2011. Macroprudential Policy Tools and Frameworks. Progress Report to G20, October.
- Galán, J.E., Lamas, M., 2019. Beyond the LTV Ratio: New Macroprudential Lessons From Spain. Working Papers No. 1931. Bank of Spain.
- Galán, J.E., Measuring Credit-to-GDP Gaps, 2019. The Hodrick-Prescott Filter Revisited. Occasional Papers No. 1906. Bank of Spain.
- Galati, G., Moessner, R., 2018. What Do We Know About the Effects of Macroprudential Policy? *Economica* 85, 735–770.
- Galati, G., Hindrayanto, I., Koopman, S.J., Vlekke, M., 2016. "Measuring financial cycles in a model-based analysis: empirical evidence for the United States and the euro area". *Econ. Lett.* 145, 83–87.
- Galvez, J., Mencía, J., 2014. Distributional Linkages between European Sovereign Bond and Bank Assets Returns. CEMFI Working Paper No. 1407. CEMFI.
- Giglio, S., Kelly, B., Pruitt, S., 2016. Systemic risk and the macroeconomy: an empirical evaluation. *J. Financ. Econ.* 119 (3), 457–471.
- Graham, B.S., Powell, J.L., 2012. Identification and estimation of average partial effects in 'irregular' correlated random coefficient panel data models. *Econometrica* 80 (5), 2105–2152.
- International Monetary Fund (IMF), 2014. Israel – Selected Issues, IMF Country Report 14/48, February.
- International Monetary Fund (IMF), 2020. Dampening Global Financial Shocks in Emerging Markets: Can Macroprudential Regulation Help?, Chapter 3 in World Economic Outlook: The Great Lockdown, World Economic Outlook Reports, April.
- Jiménez, G., Ongena, S., Peydró, J.L., Saurina, J., 2017. Macroprudential policy, countercyclical bank capital buffers, and credit supply: evidence from the Spanish dynamic provisioning experiments. *J. Polit. Econ.* 125 (6), 2126–2177.
- Jordà, O., Schularick, M., Taylor, A.M., 2015. Leveraged bubbles. *J. Monet. Econ.* 76, 1–20.
- Jorion, P., 2001. Value at Risk – the New Benchmark for Managing Financial Risk. McGraw-Hill, Chicago.
- Kato, K., Galvão, A.F., Montes-Rojas, G., 2012. Asymptotics for panel quantile regression models with individual effects. *J. Econom.* 170, 76–91.
- Kim, S., Mehrotra, A., 2018. Effects of monetary and macroprudential policies—evidence from four inflation targeting economies. *J. Money Credit Bank.* 50 (5), 967–992.
- Koenker, R., 2004. Quantile regression for longitudinal data. *J. Multivar. Anal.* 91, 74–89.
- Koenker, R., 2005. Quantile Regression. Cambridge University Press, Cambridge.
- Koenker, R., Bassett, G., 1978. Regression quantiles. *Econometrica* 46 (1), 33–50.
- Koenker, R., Geling, O., 2001. Reappraising medfly longevity: a quantile regression approach. *J. Comput. Graph. Stat.* 96, 458–468.
- Koenker, R., Machado, J.A.F., 1999. Goodness of fit and related inference processes for quantile regression. *J. Am. Stat. Assoc.* 94 (448), 1296–1310.
- Kuttner, K.N., Shim, I., 2016. Can non-interest rate policies stabilize housing markets? Evidence from a panel of 57 economies. *J. Financ. Stab.* 26, 31–44.
- Laeven, L., Valencia, F., 2018. Systemic Banking Crises Revisited. IMF Working Paper WP/18/123. International Monetary Fund.
- Lang, J.H., Izzo, C., Fahr, S., Ruzicka, J., 2019. Anticipating the Bust: a New Cyclical Systemic Risk Indicator to Assess the Likelihood and Severity of Financial Crises. Occasional Paper Series 219. European Central Bank.
- Lim, C.H., Columba, F., Costa, A., Kongsamut, P., Otani, A., Saiyid, M., Wezel, T., Wu, X., 2011. Macroprudential Policy: What Instruments and How Are They Used? Lessons from Country Experiences. IMF Working Paper 11/238. International Monetary Fund.
- Loria, F., Mathhes, C., Zhang, D., 2019. Assessing Macroeconomic Tail Risk. Working Paper Series 1910. Federal Reserve Bank of Richmond.
- Machado, J.A.F., Santos Silva, J.M.C., 2019. Quantiles via moments. *J. Econom.* 213 (1), 145–173.
- Noss, J., Toffano, P., 2016. Estimating the impact of changes in aggregate bank capital requirements on lending and growth during an upswing. *J. Bank. Financ.* 62, 15–27.
- Poghosyan, T., 2019. "How Effective Is Macroprudential Policy? Evidence from Lending Restriction Measures in EU Countries". IMF Working Paper 19/45. International Monetary Fund.
- Richter, B., Schularik, M., Shim, I., 2019. The costs of macroprudential policy. *J. Int. Econ.* 118, 263–282.